还原和恢复 SAP HANA 数据库

Snap Creator Framework

NetApp
January 31, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/snap-creator-framework/sap-hana-ops/task_restoring_and_recovering_databases_from_primary_storage.html on January 31, 2022. Always check docs.netapp.com for the latest.

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还原和恢复 SAP HANA 数据库

您可以使用 SAP HANA Studio 和 Snap Creator 还原和恢复 SAP HANA 数据库。

1. 在 SAP HANA Studio 中：
   a. 为 SAP HANA 系统选择恢复。
   b. SAP HANA 系统已关闭。
   c. 选择恢复类型。
   d. 提供日志备份位置。
   e. 此时将显示数据备份列表
   f. 选择 backup 可查看外部备份 ID 。
2. 对于仅运行集群模式 Data ONTAP 的存储系统：
   a. 只有在使用最新备份以外的任何其他备份进行还原时才需要此备份。
   b. 仅主存储中的 "`Volume SnapRestore` " 才需要此功能。
   c. 停用 SnapVault 关系
3. 在 Snap Creator 中：
   a. 为 SAP HANA 系统选择 "`Restore` " 。
   b. 根据主存储备份的可用性，选择从主存储或二级存储还原。
   c. 选择存储控制器，卷名称和 Snapshot 副本名称。Snapshot 副本名称与先前的备份 ID 相关。
   d. 对于多节点 SAP HANA 系统，需要还原多个卷：
      i. 选择 * 添加更多还原项 * 。
      ii. 选择存储控制器，卷名称和 Snapshot 副本名称。
      iii. 对所有所需卷重复此过程。
   e. 对于多租户数据库容器（ MDC ）单租户数据库系统，系统和租户数据库都将还原。
   f. 还原过程已启动
   g. 已完成所有卷的还原。
4. 在数据库节点上，卸载并挂载所有数据卷以清理 " 陈旧的 NFS 句柄 " 。
5. 在 SAP HANA Studio 中：
   a. 在备份列表中选择 * 刷新 * 。
   b. 选择可用于恢复的备份（绿色项目）。
   c. 启动恢复过程。
   d. 对于多租户数据库容器（ MDC ）单租户数据库系统，首先为系统数据库启动恢复过程，然后为租户数据库启动恢复过程。
   e. SAP HANA 系统已启动。
6. （可选）恢复所有已还原卷的 SnapVault 关系。

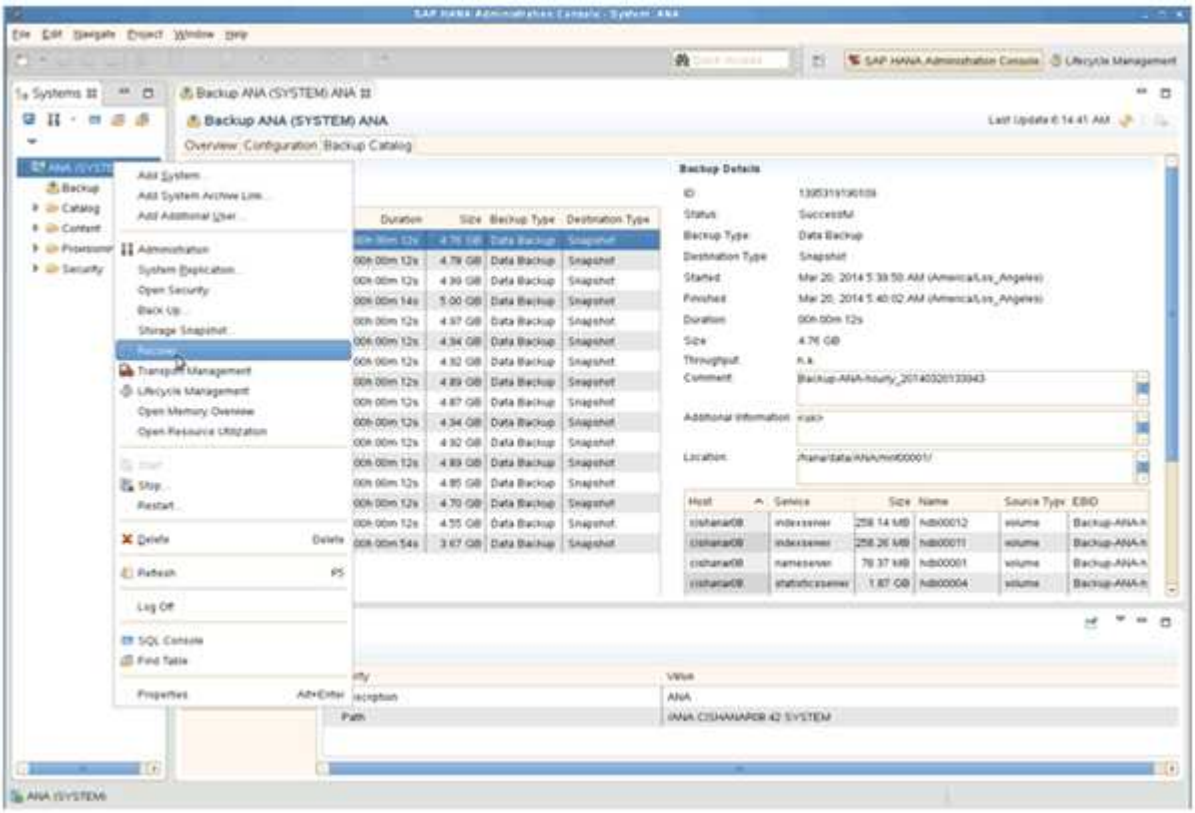
在存储系统中，只有在使用最新备份以外的备份进行还原时，才需要执行此步骤。

从主存储还原和恢复数据库

您可以从主存储还原和恢复数据库。

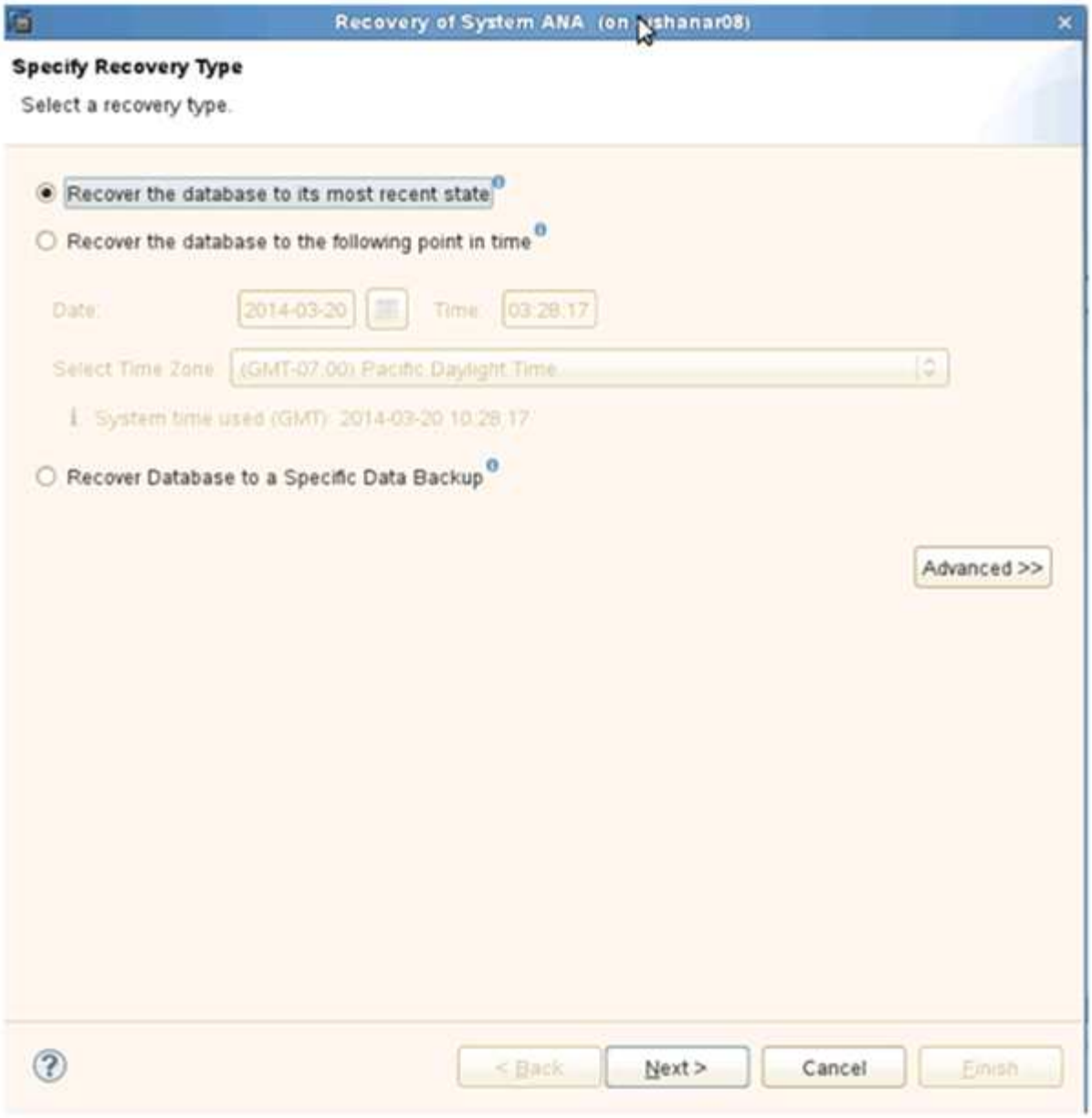
您不能从 Snap Creator 还原基于文件的备份副本。

1. 在 SAP HANA Studio 中，为 SAP HANA 系统选择 * 恢复 * 。

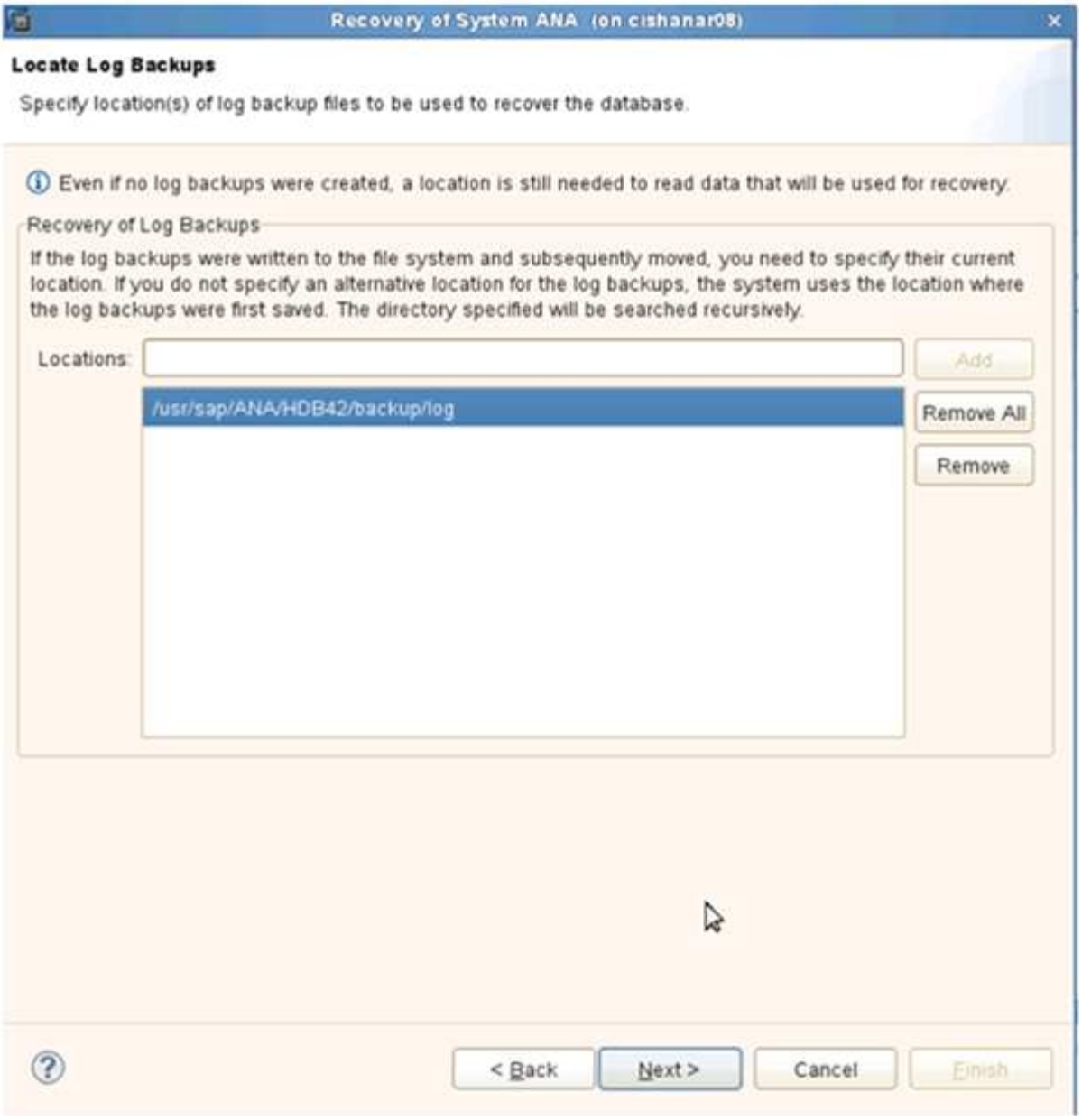


SAP HANA 系统关闭。

2. 选择恢复类型并单击 * 下一步 * 。

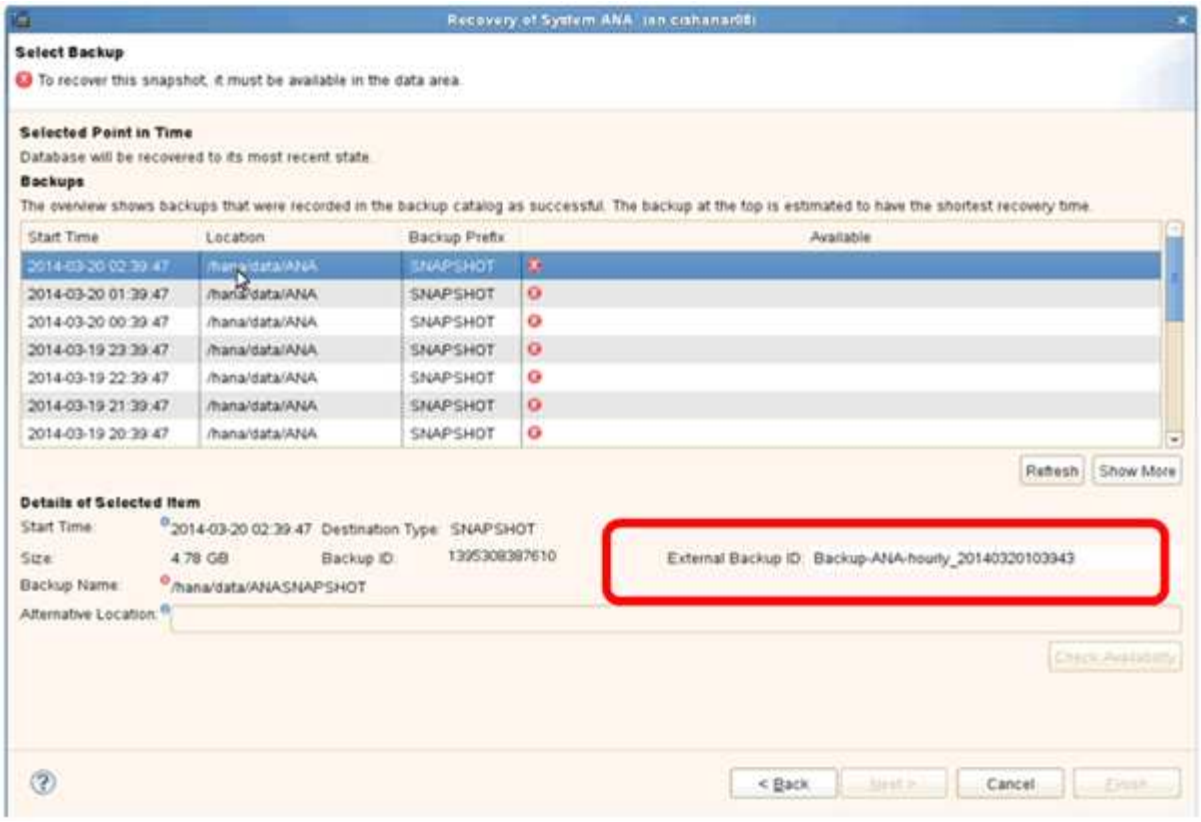


3. 提供日志备份位置，然后单击 * 下一步 * 。



您看到的可用备份列表取决于备份目录的内容。

4. 选择所需的备份并记录外部备份 ID 。



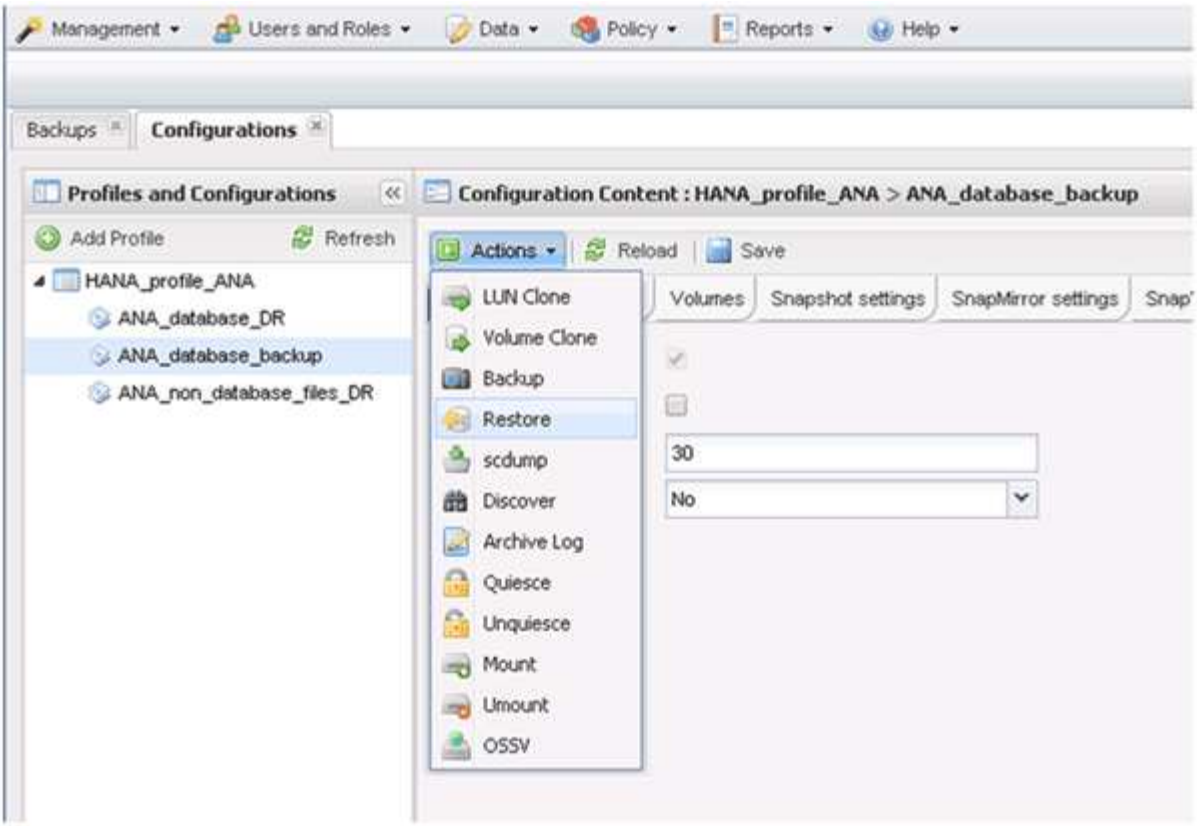
5. 停用 SnapVault 关系。

> 只有集群模式 Data ONTAP 才需要执行此步骤。

如果需要还原的 Snapshot 副本早于当前用作 SnapVault 基本 Snapshot 副本的 Snapshot 副本，则必须先在集群模式 Data ONTAP 中停用 SnapVault 关系。为此，请在备份集群控制台上执行以下命令：

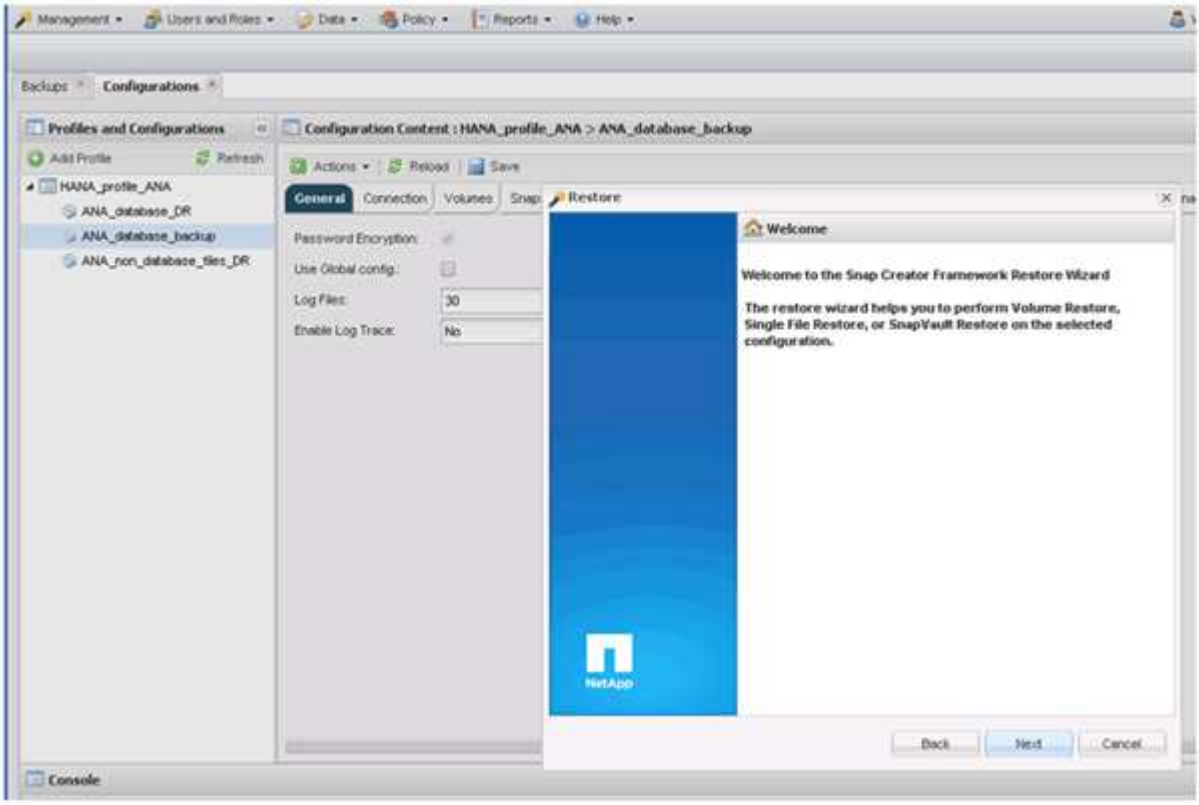
```
hana::> snapmirror quiesce -destination-path hana2b:backup_hana_data
Operation succeeded: snapmirror quiesce for destination
hana2b:backup_hana_data.

hana::> snapmirror delete -destination-path hana2b:backup_hana_data
Operation succeeded: snapmirror delete the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror release -destination-path hana2b:backup_hana_data
[Job 6551] Job succeeded: SnapMirror Release Succeeded
```

6. 在 Snap Creator GUI 中，选择 SAP HANA 系统，然后选择 * 操作 * > * 还原 * 。

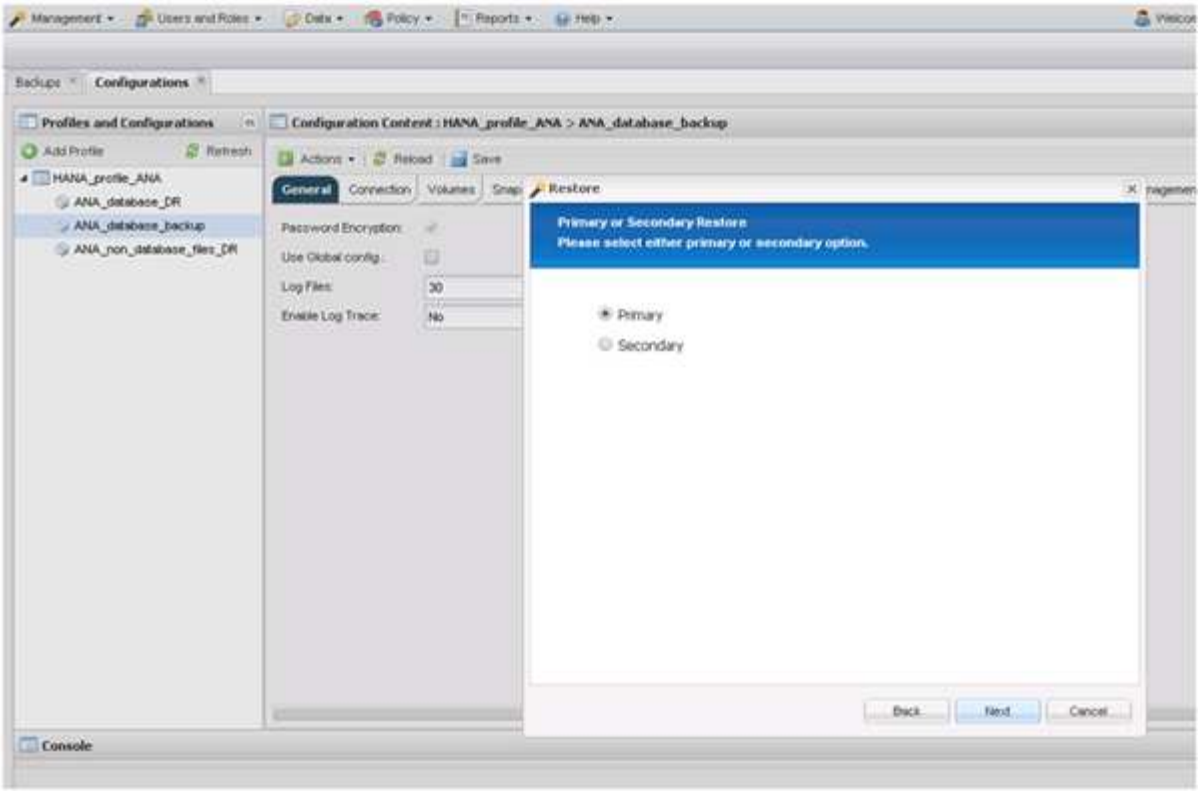


此时将显示 Welcome to the Snap Creator Framework Restore Wizard 屏幕。

7. 单击 * 下一步 * 。

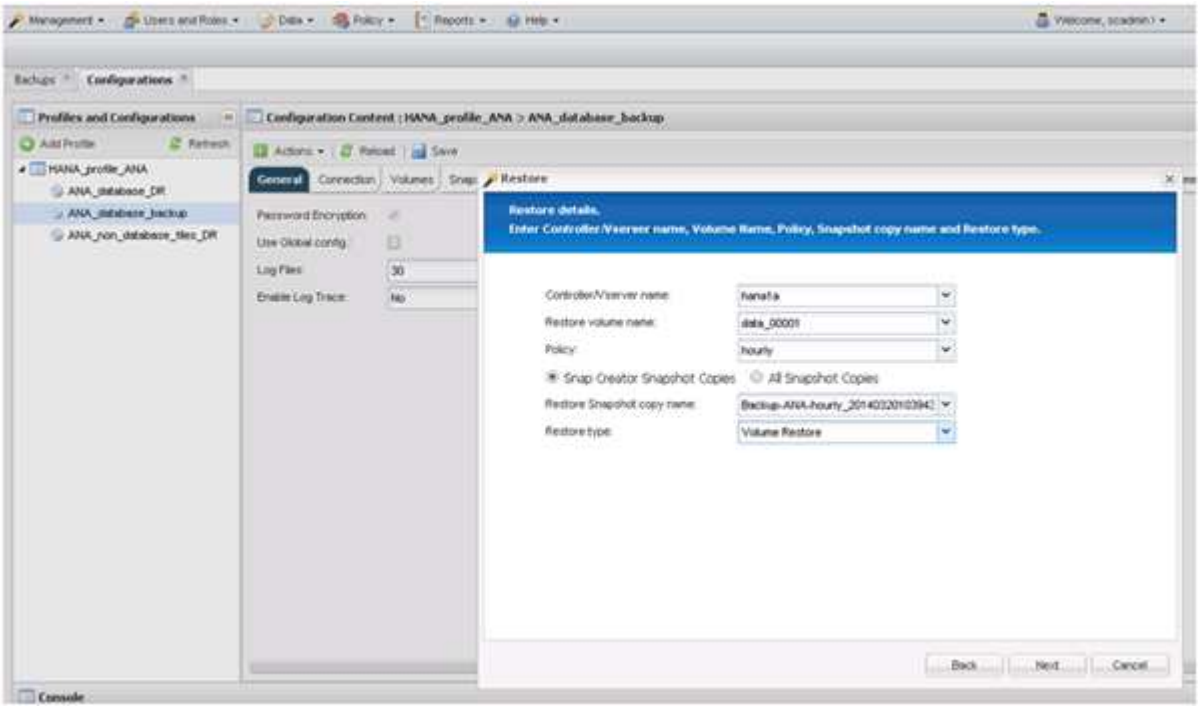


8. 选择 * 主 * 并单击 * 下一步 * 。

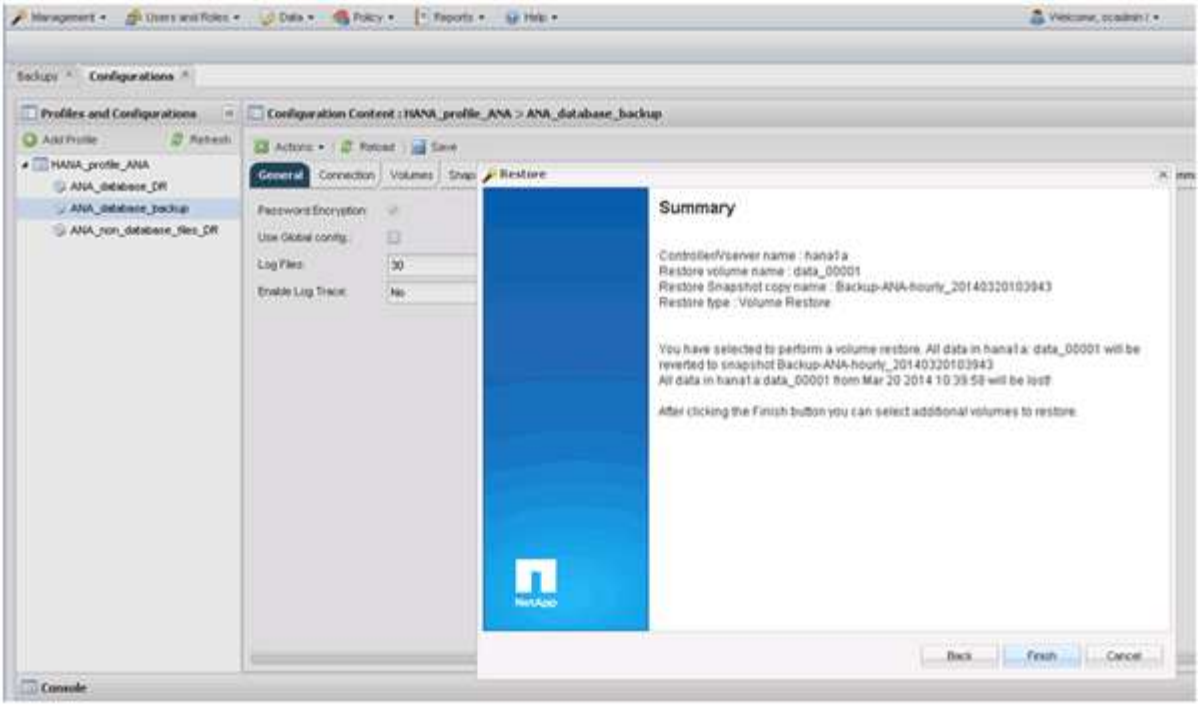


9. 选择从主存储还原。
10. 选择存储控制器，卷名称和 Snapshot 名称。

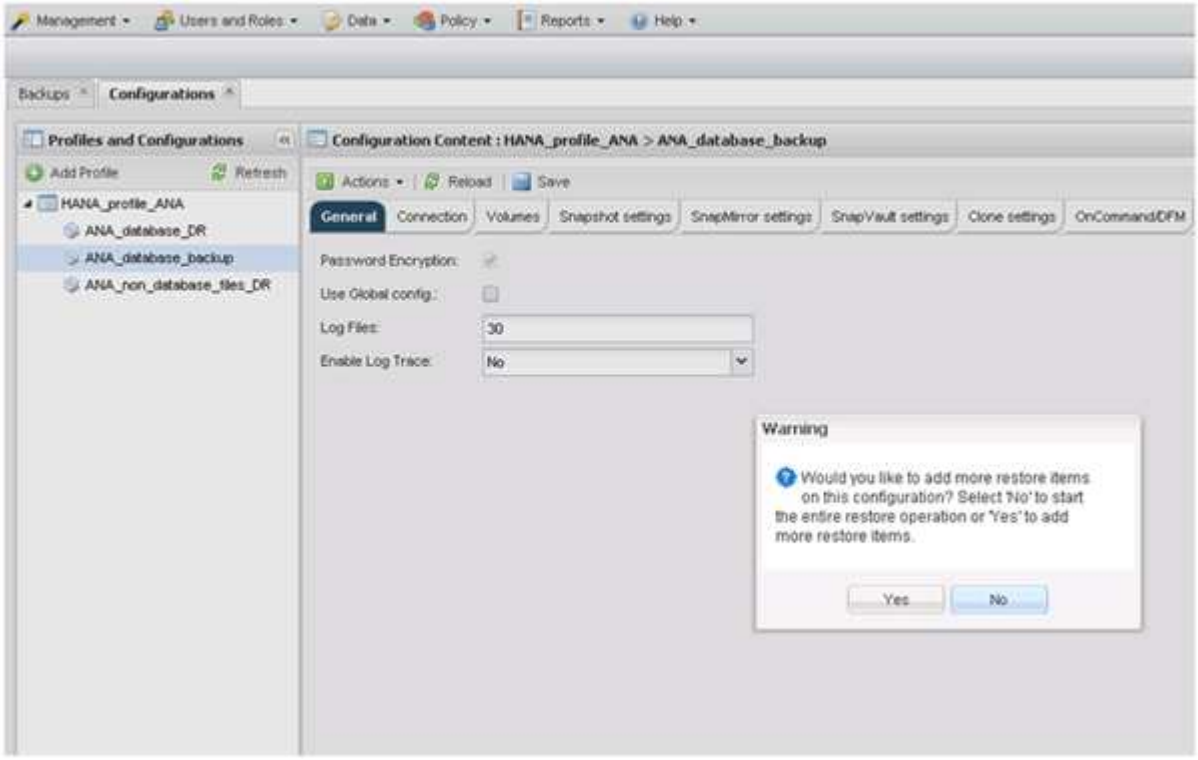
    Snapshot 名称与在 SAP HANA Studio 中选择的备份 ID 关联。



11. 单击 * 完成 * 。

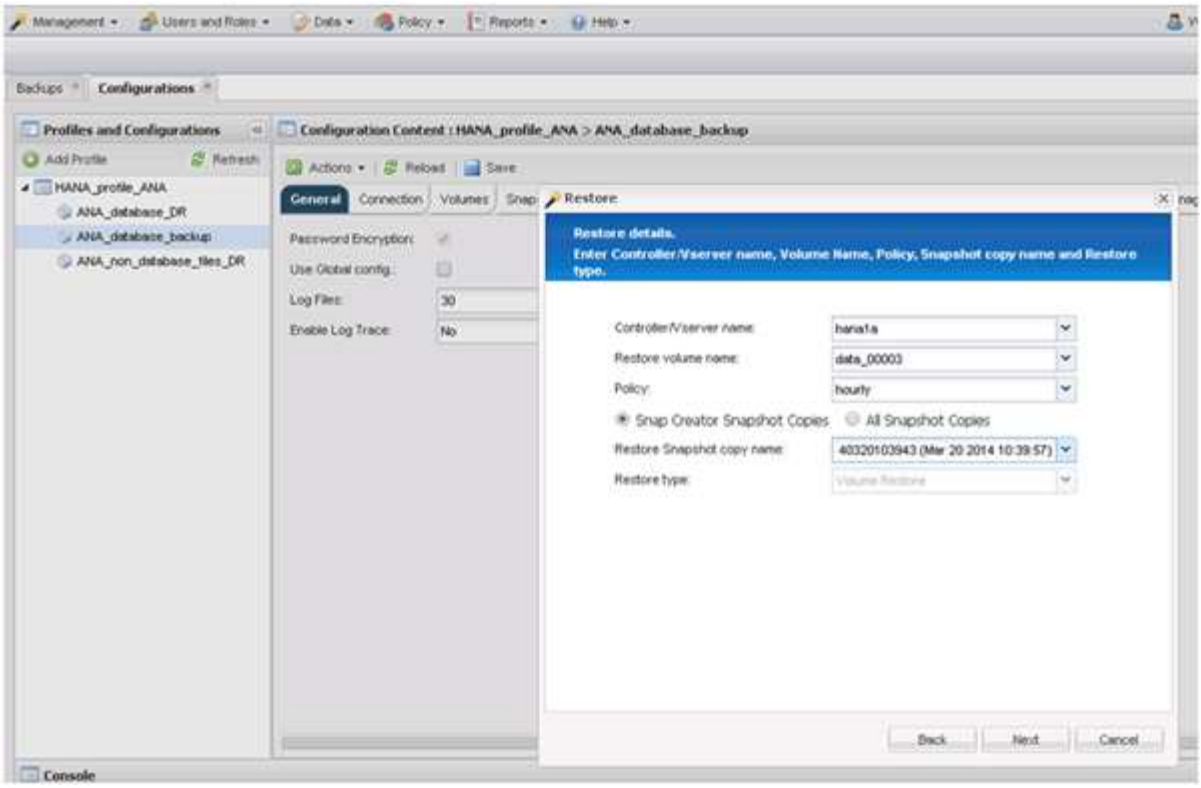


12. 单击 * 是 * 添加更多还原项。

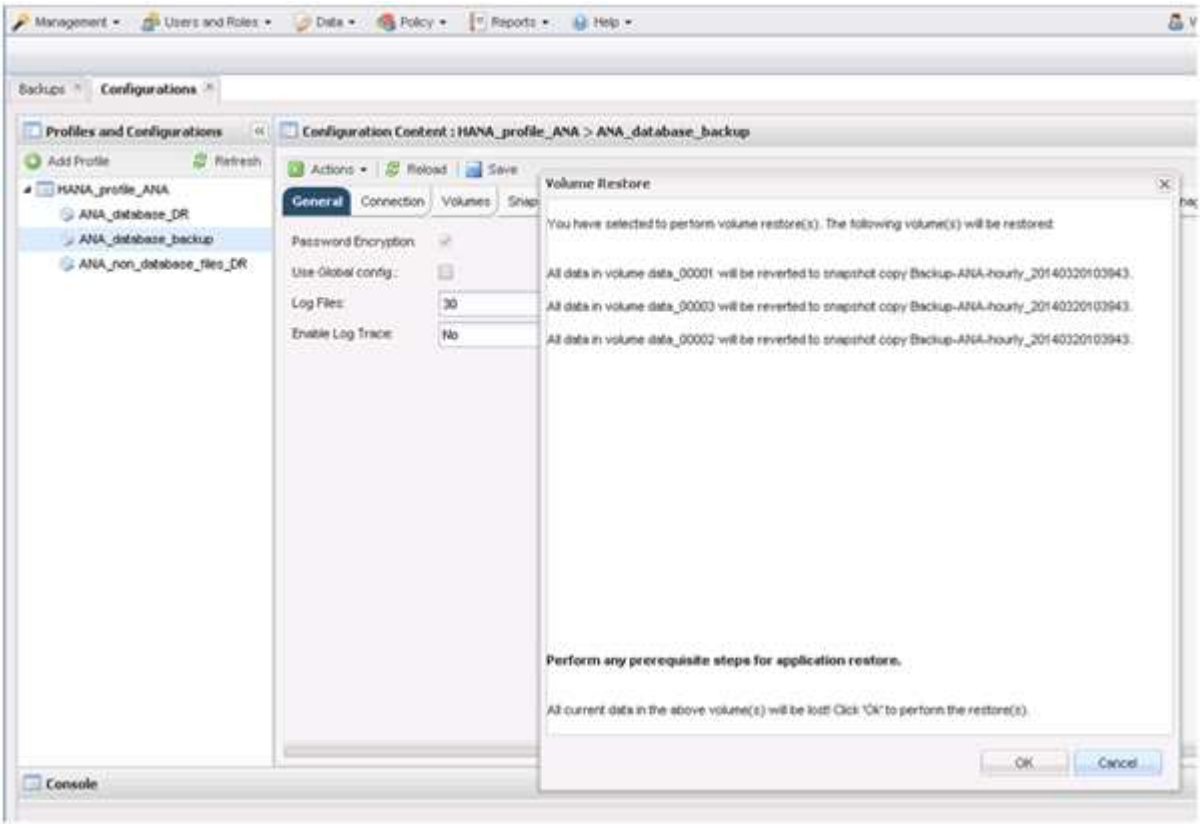


13. 选择存储控制器，其他卷名称和 Snapshot 名称。

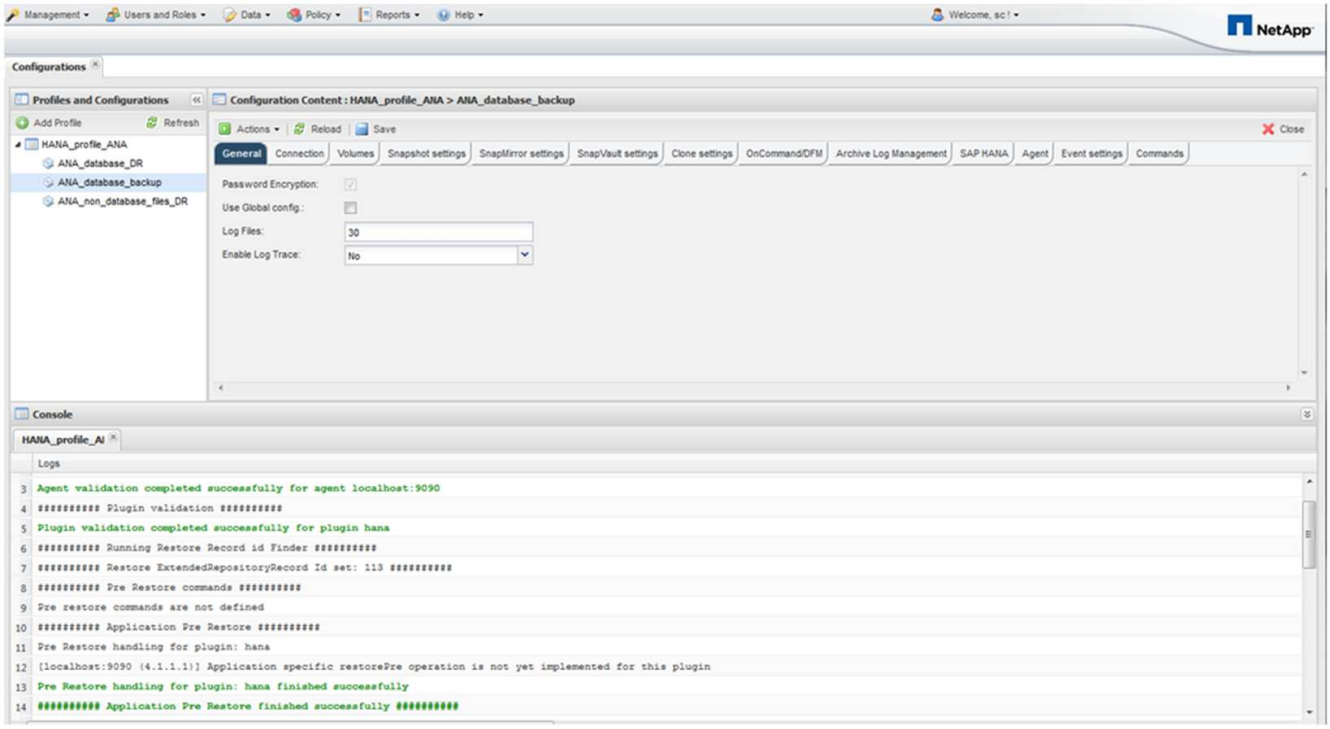
    Snapshot 名称与在 SAP HANA Studio 中选择的备份 ID 关联。



14. 重复步骤 10 到 13 ，直到添加了所有必需的卷；在本示例中，需要为还原过程选择 data_00001 ，data_00002 和 data_00003 。
15. 选择所有卷后，单击 * 确定 * 以启动还原过程。



此时将启动还原过程。



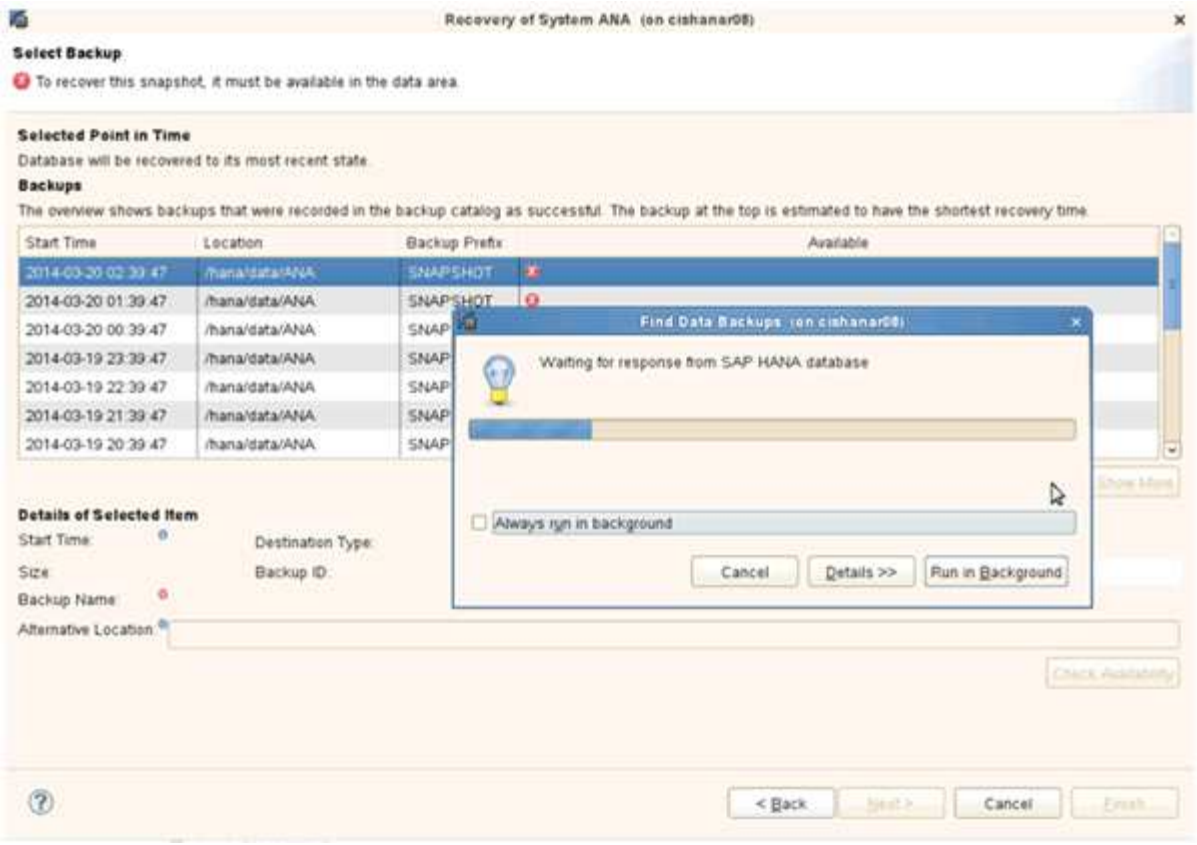
请等待还原过程完成。

16. 在每个数据库节点上，重新挂载所有数据卷以清除陈旧的 NFS 句柄。

    在此示例中，所有三个卷都需要在每个数据库节点上重新挂载。

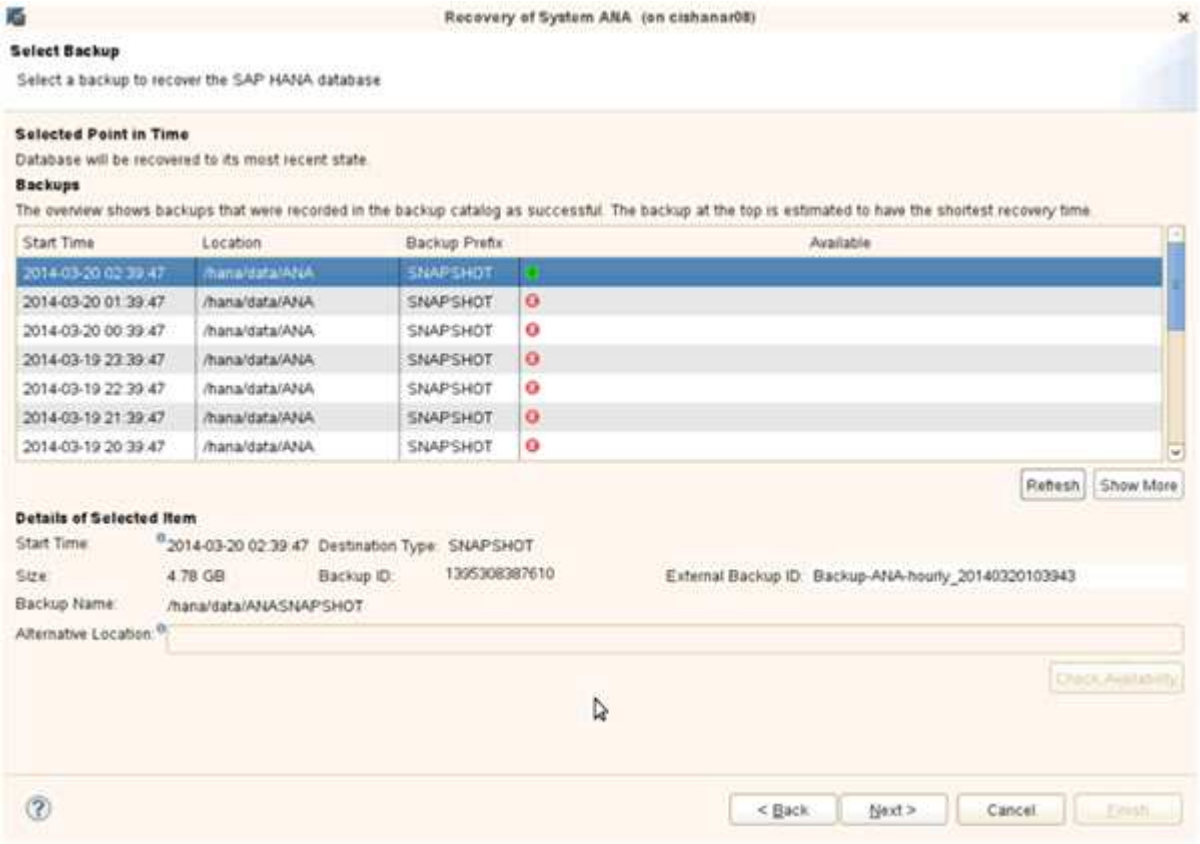
```
mount -o remount /hana/data/ANA/mnt00001
mount -o remount /hana/data/ANA/mnt00002
mount -o remount /hana/data/ANA/mnt00003
```

17. 转到 SAP HANA Studio 并单击 * 刷新 * 以更新可用备份列表。

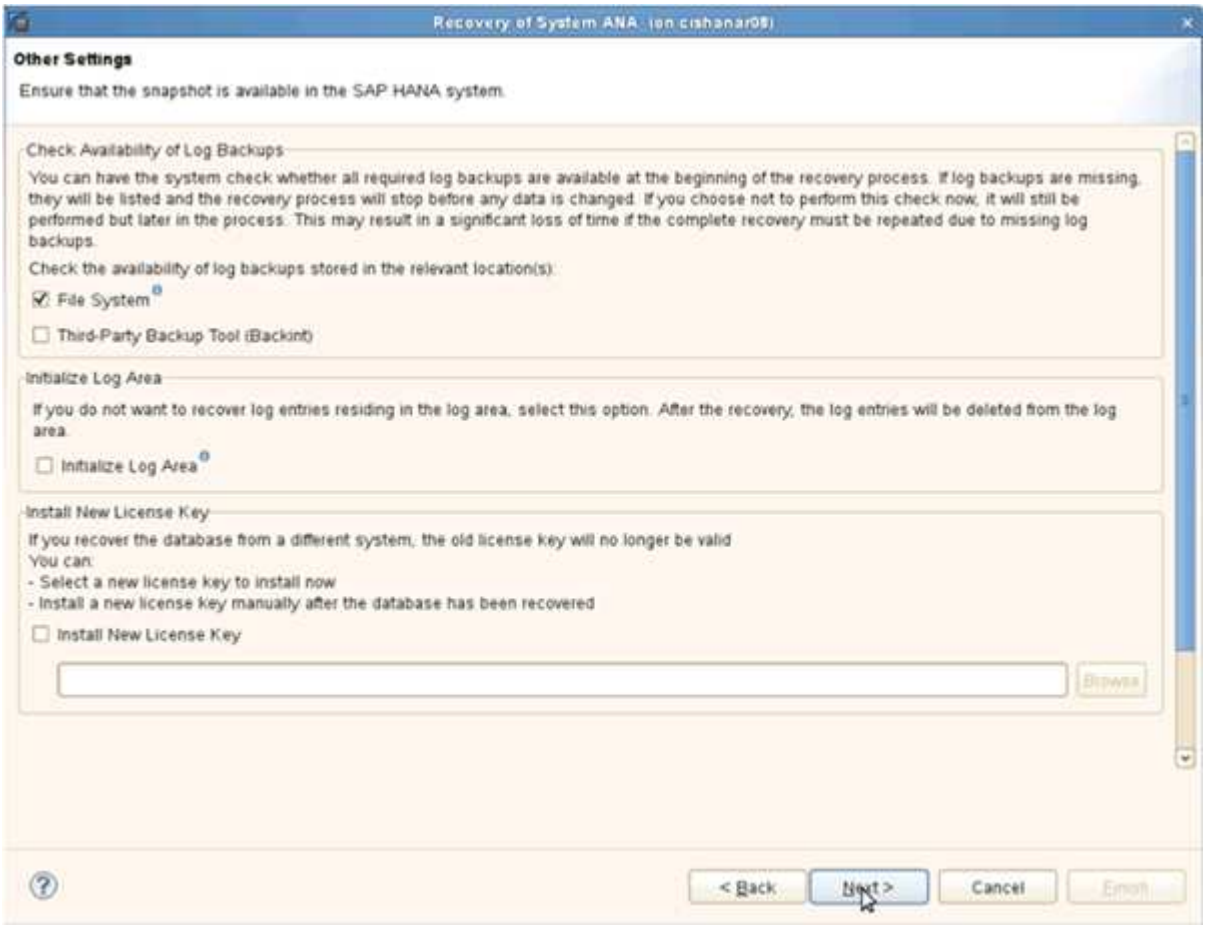


使用 Snap Creator 还原的备份在备份列表中显示绿色图标。

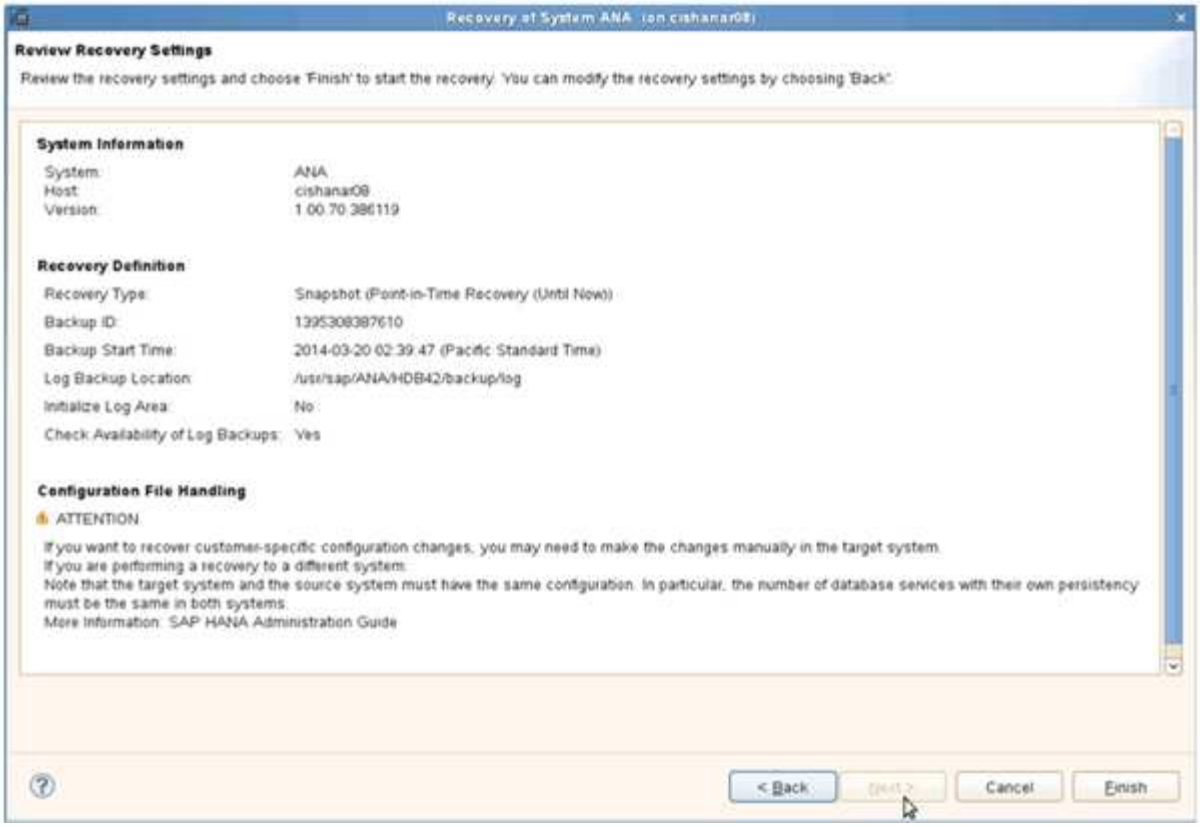
18. 选择备份并单击 * 下一步 * 。



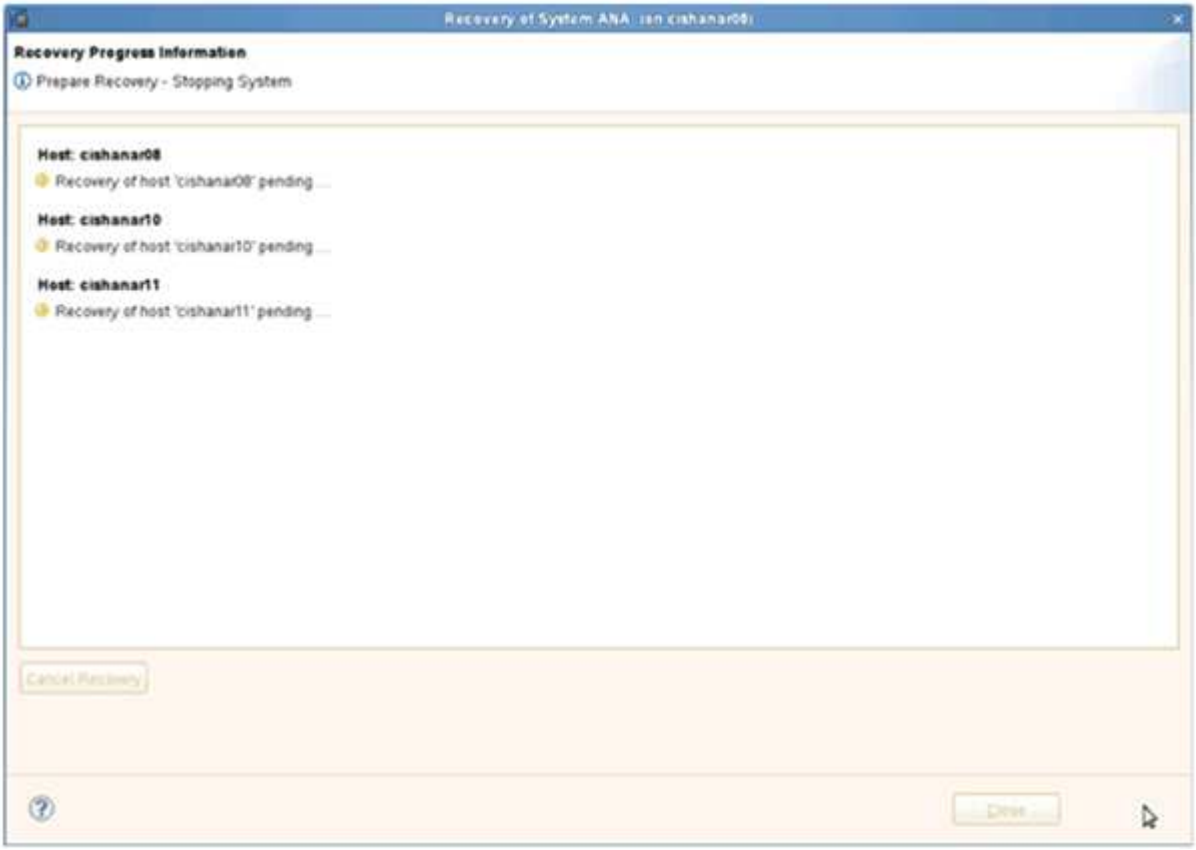
19. 根据需要选择其他设置，然后单击 * 下一步 * 。



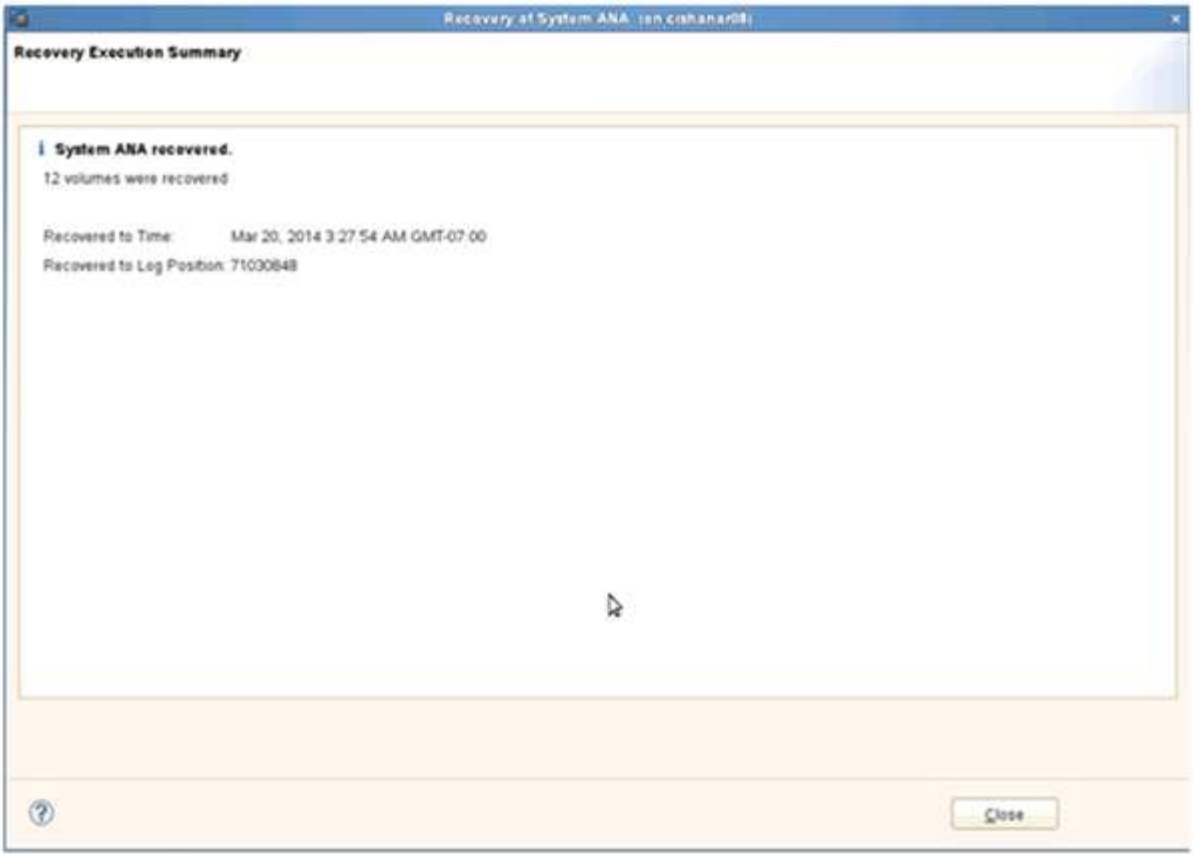
20. 单击 * 完成 * 。



恢复过程开始。



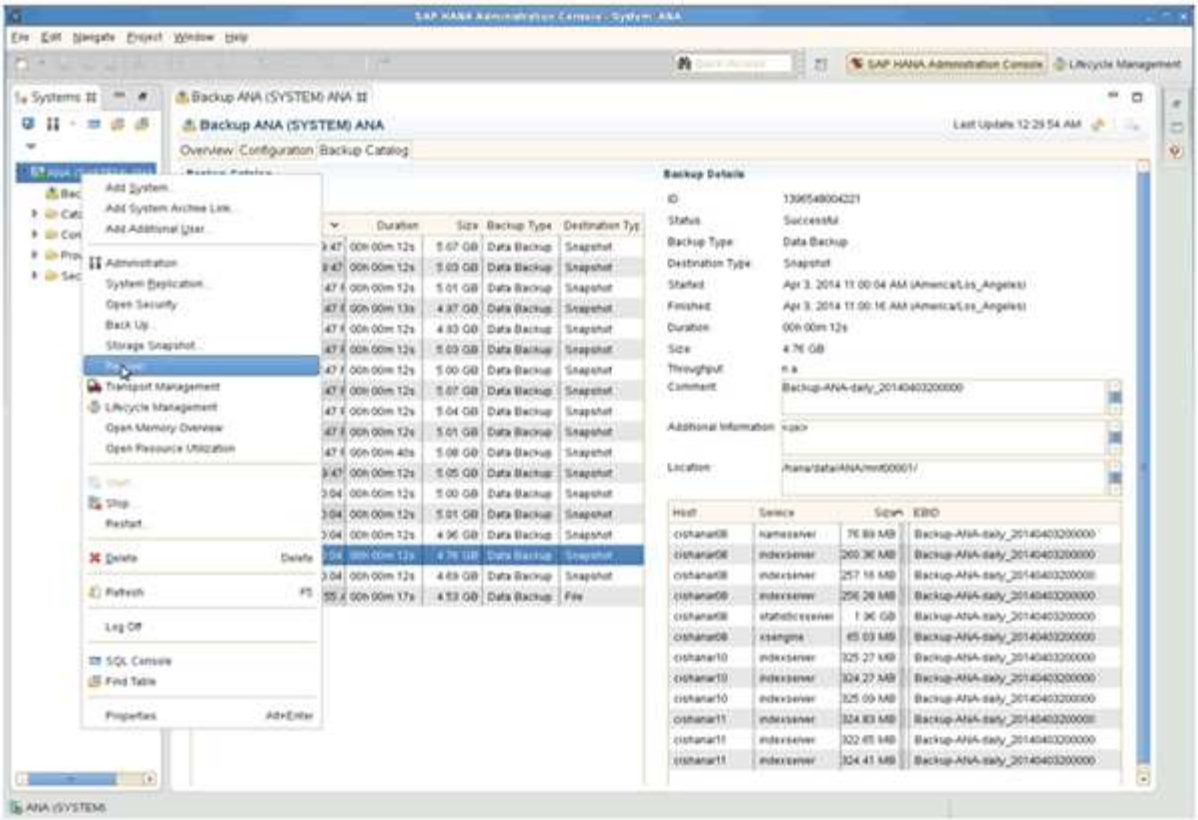
21. 恢复完成后，根据需要恢复 SnapVault 关系。



从二级存储还原和恢复数据库

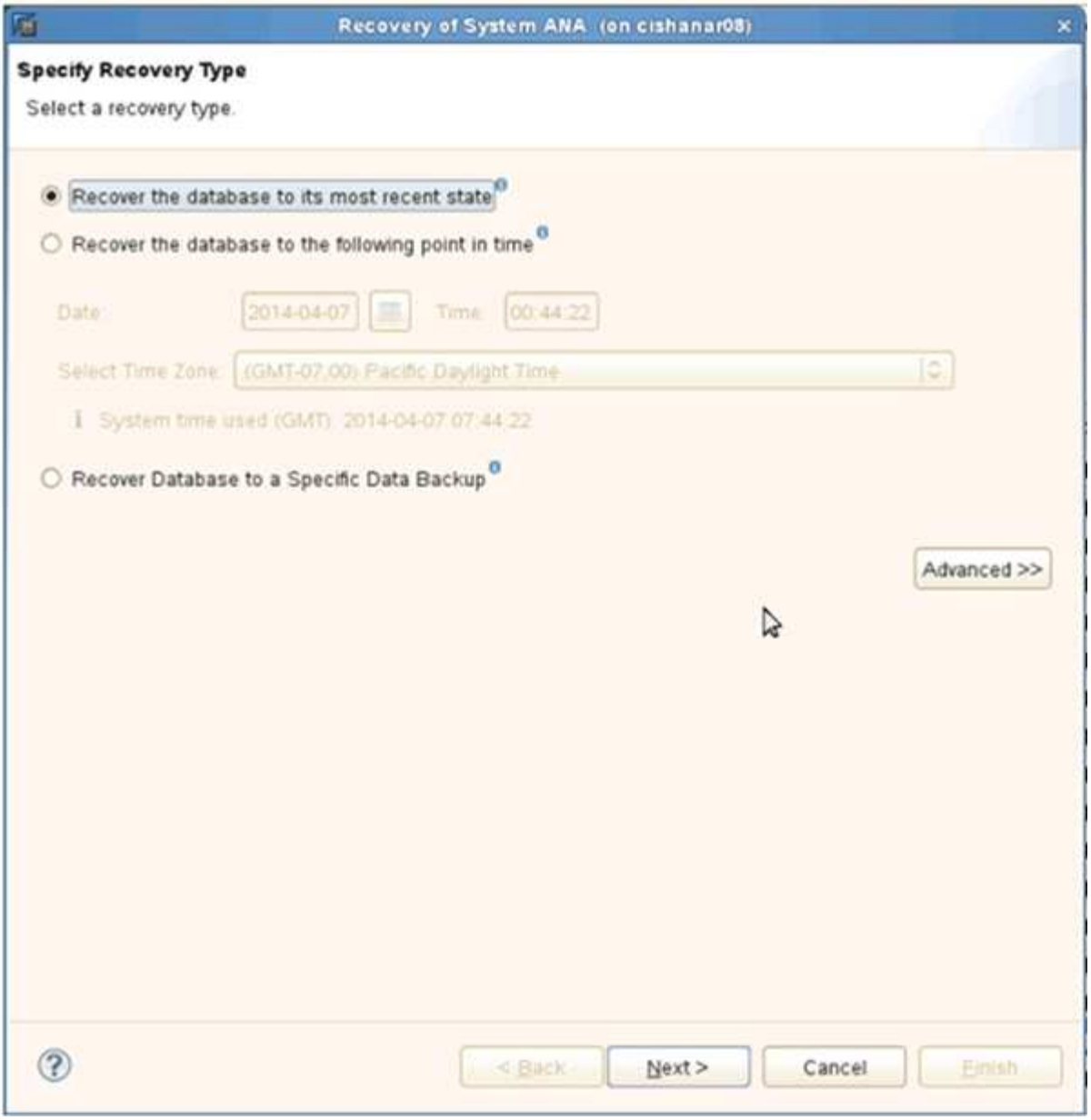
您可以从二级存储还原和恢复数据库。

1. 在 SAP HANA Studio 中，为 SAP HANA 系统选择 * 恢复 * 。

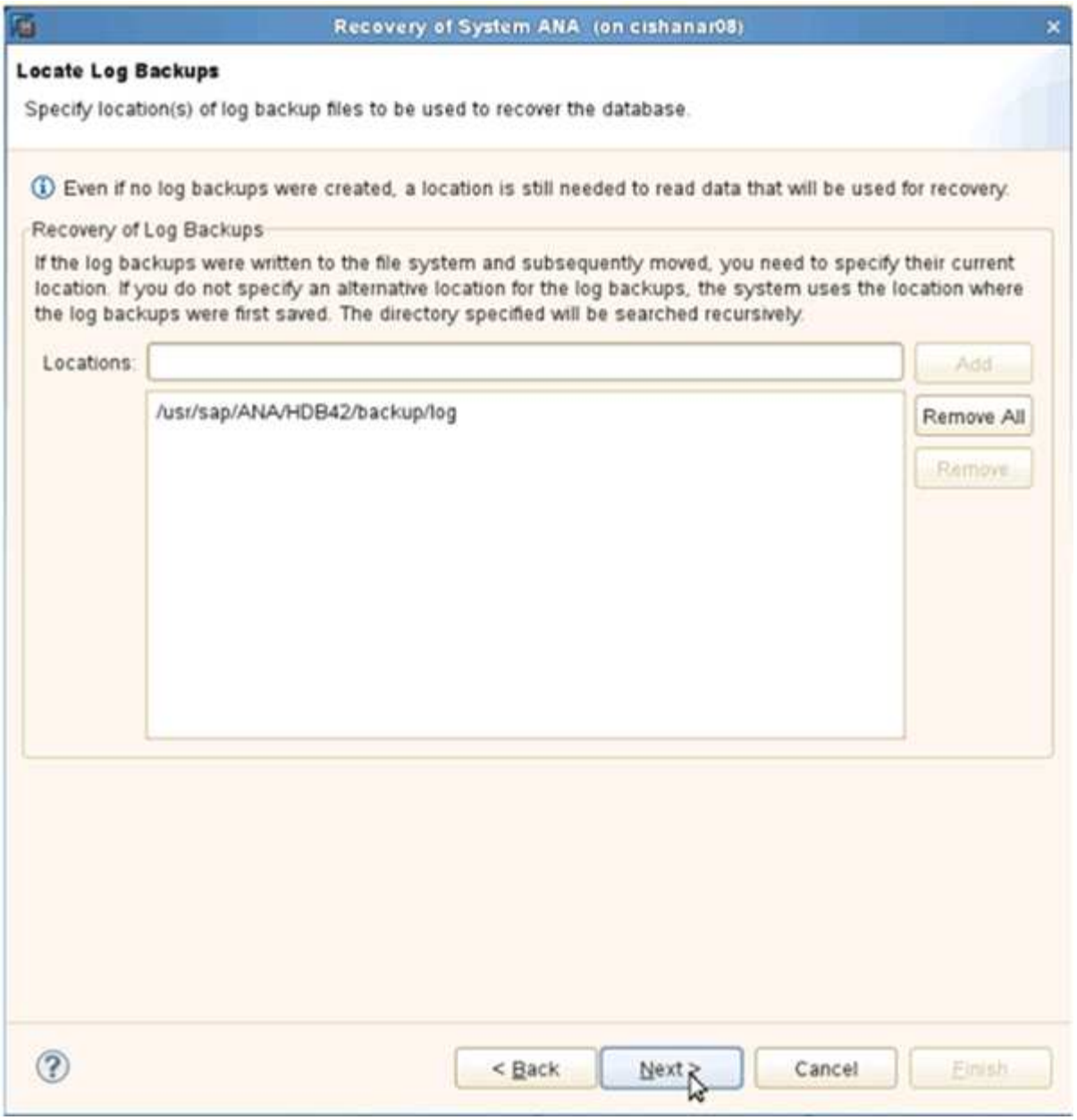


SAP HANA 系统将关闭。

2. 选择恢复类型并单击 * 下一步 * 。

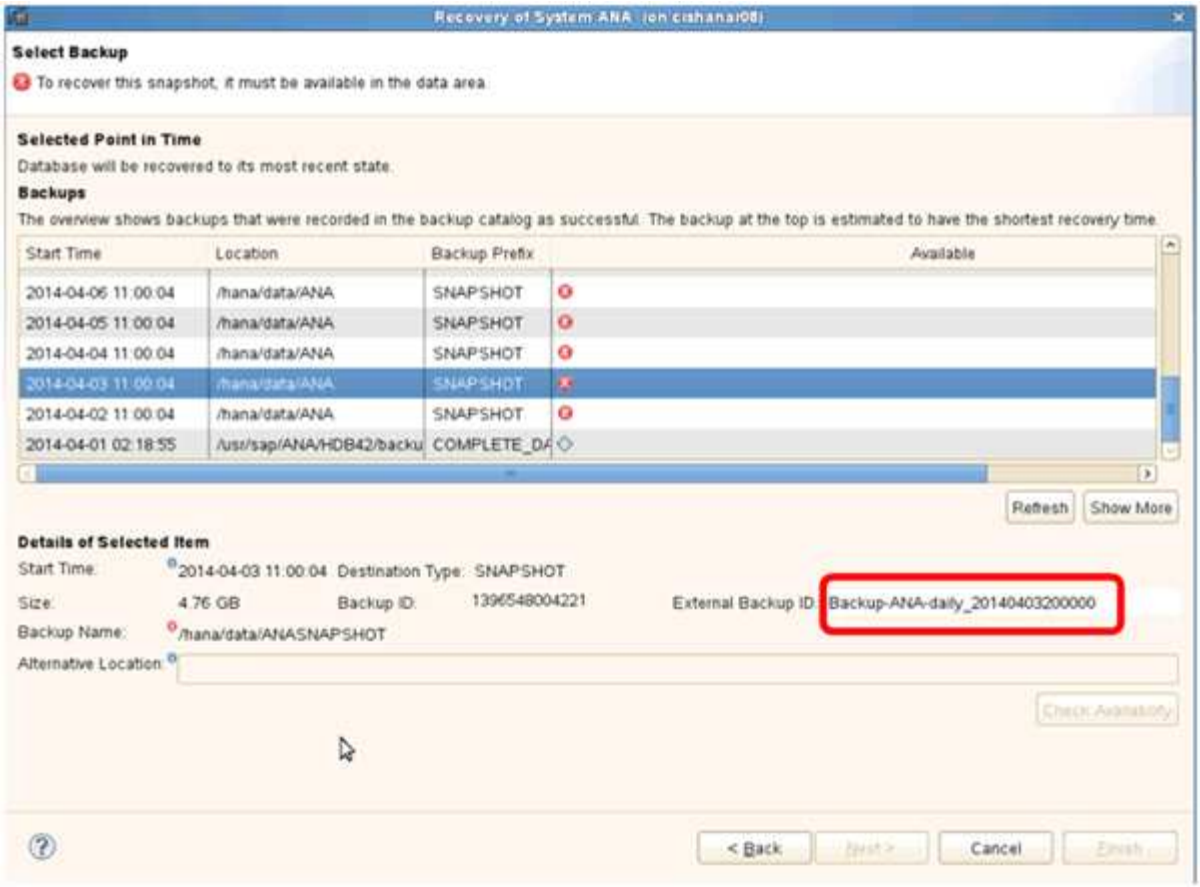


3. 提供日志备份位置，然后单击 * 下一步 * 。

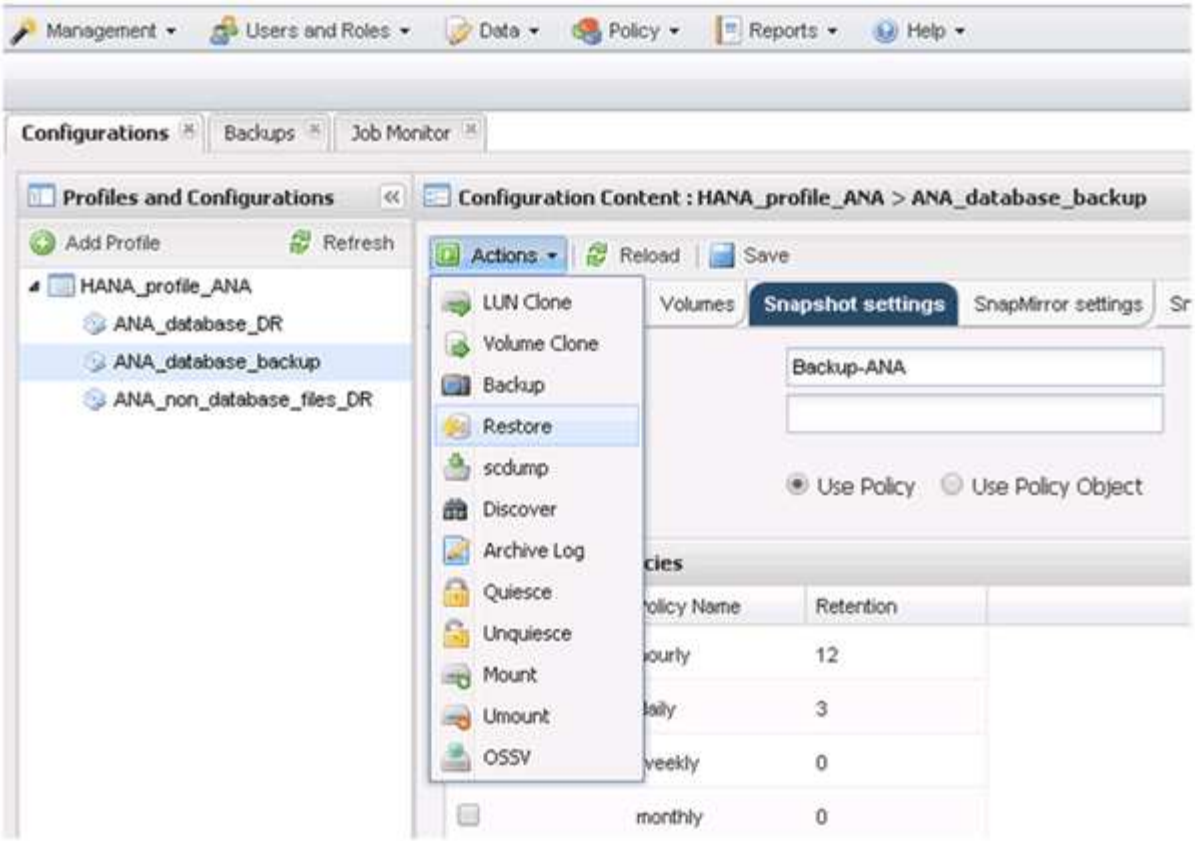


此时将根据备份目录的内容显示可用备份列表。

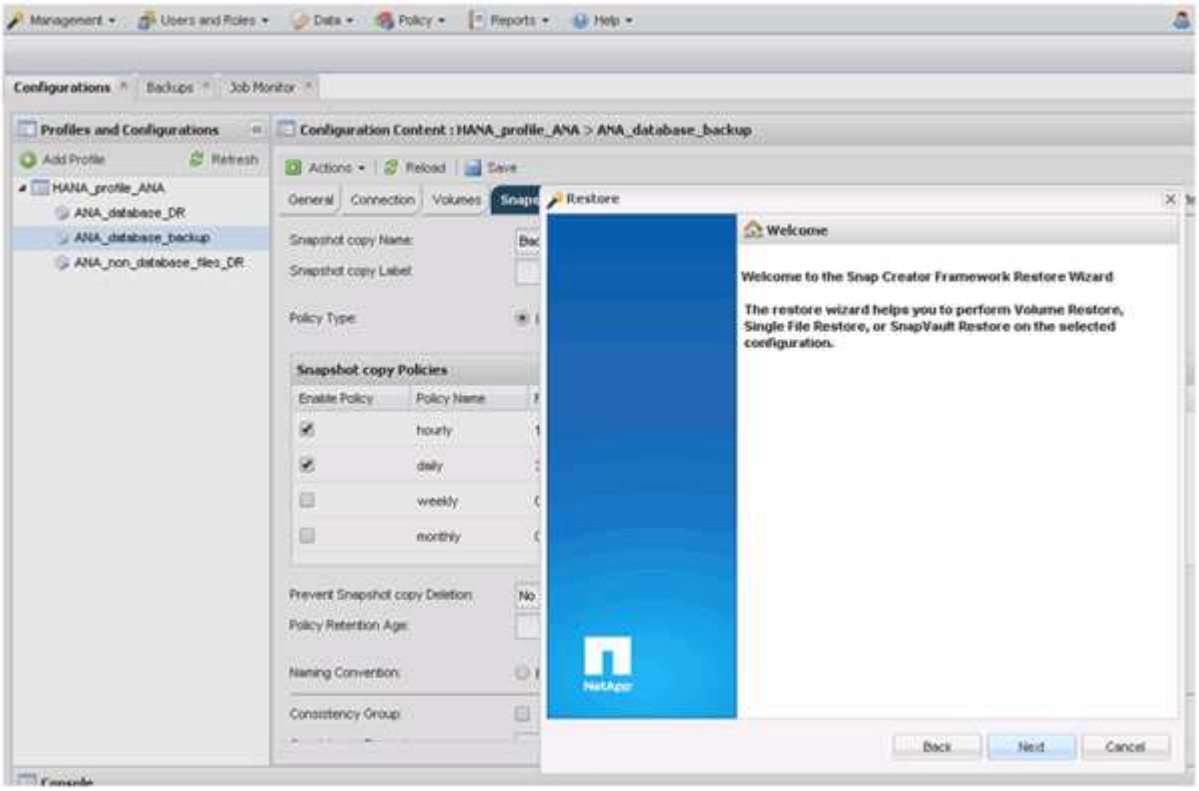
4. 选择所需的备份并记下外部备份 ID 。



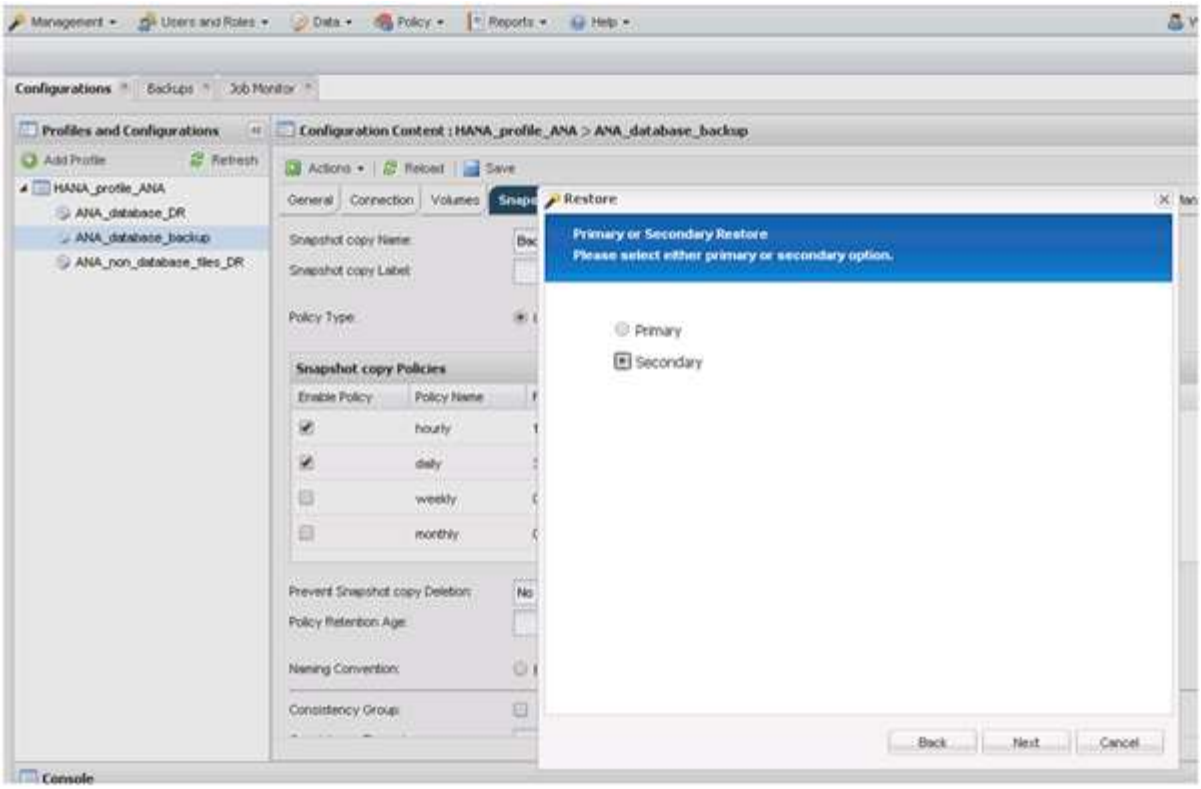
5. 转到 Snap Creator GUI 。
6. 选择 SAP HANA 系统，然后单击 * 操作 * > * 还原 * 。



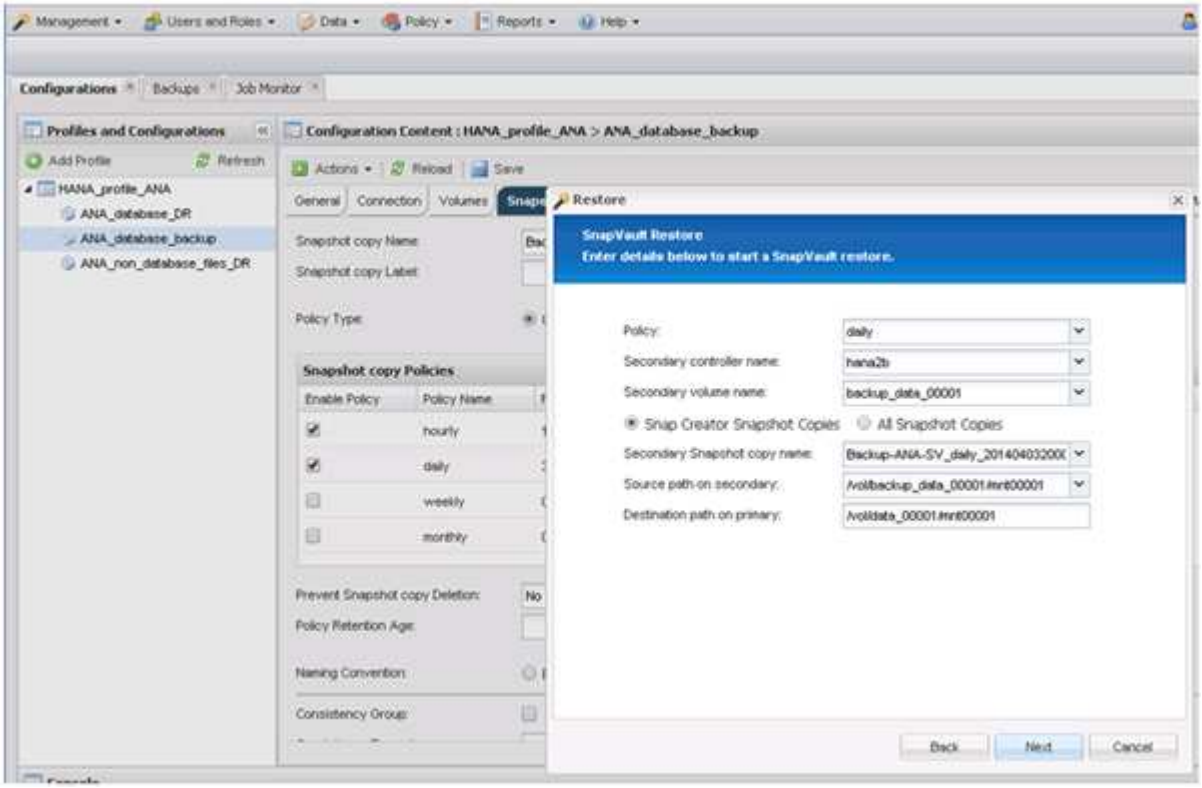
此时将显示欢迎屏幕。



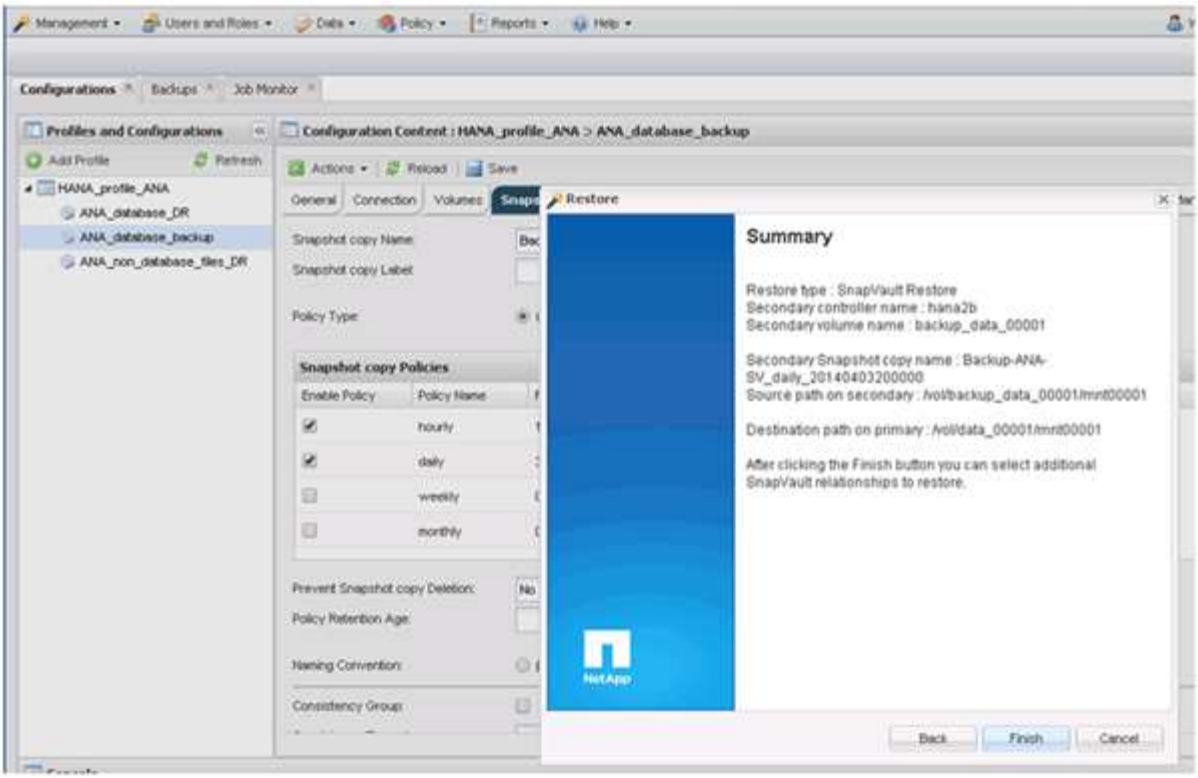
7. 单击 * 下一步 * 。
8. 选择 * 二级 * 并单击 * 下一步 * 。



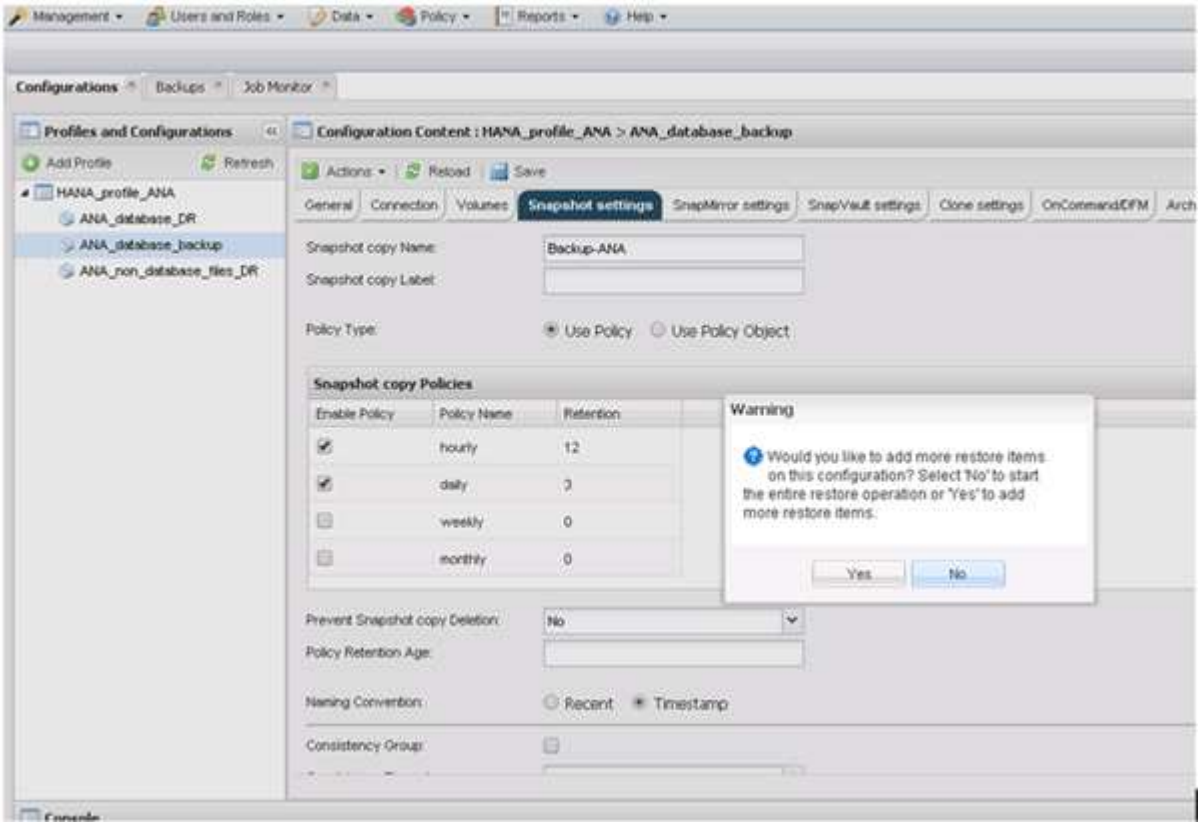
9. 输入所需信息。Snapshot 名称与在 SAP HANA Studio 中选择的备份 ID 关联。



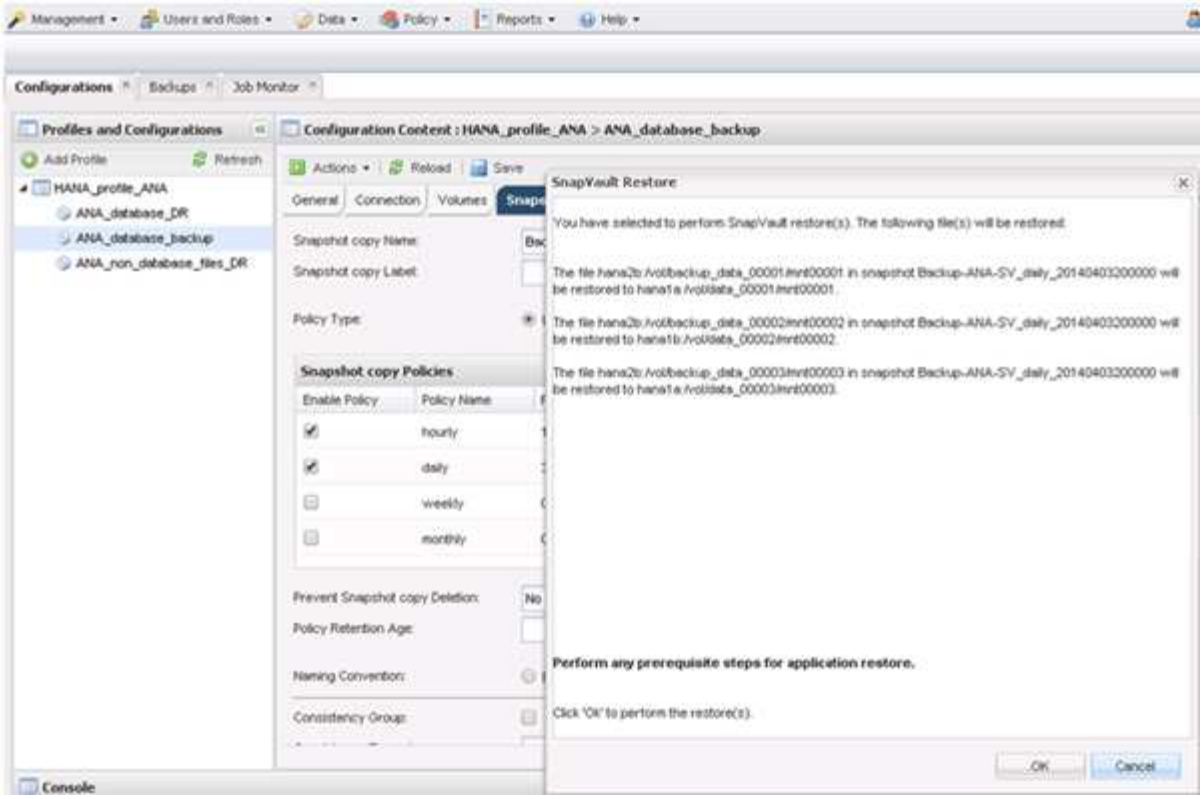
10. 选择 * 完成 * 。



11. 单击 * 是 * 添加更多项以进行还原。



12. 为需要还原的所有卷提供所需信息。在设置 data_00001 中，需要为还原过程选择 data_00002 和 data_00003 。



13. 选择所有卷后，选择 * 确定 * 以启动还原过程。

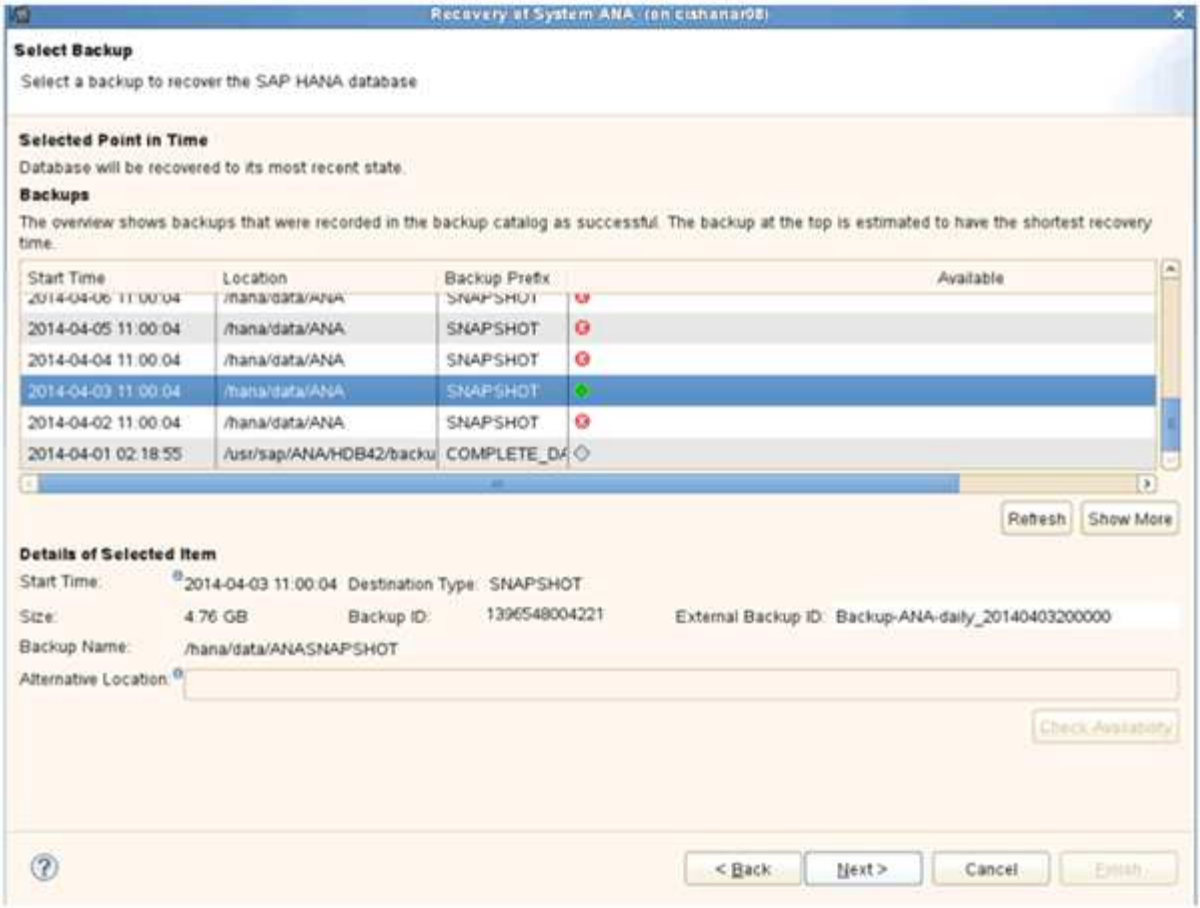
    请等待还原过程完成。

14. 在每个数据库节点上，重新挂载所有数据卷以清理 " `SNFS` 句柄 " 。

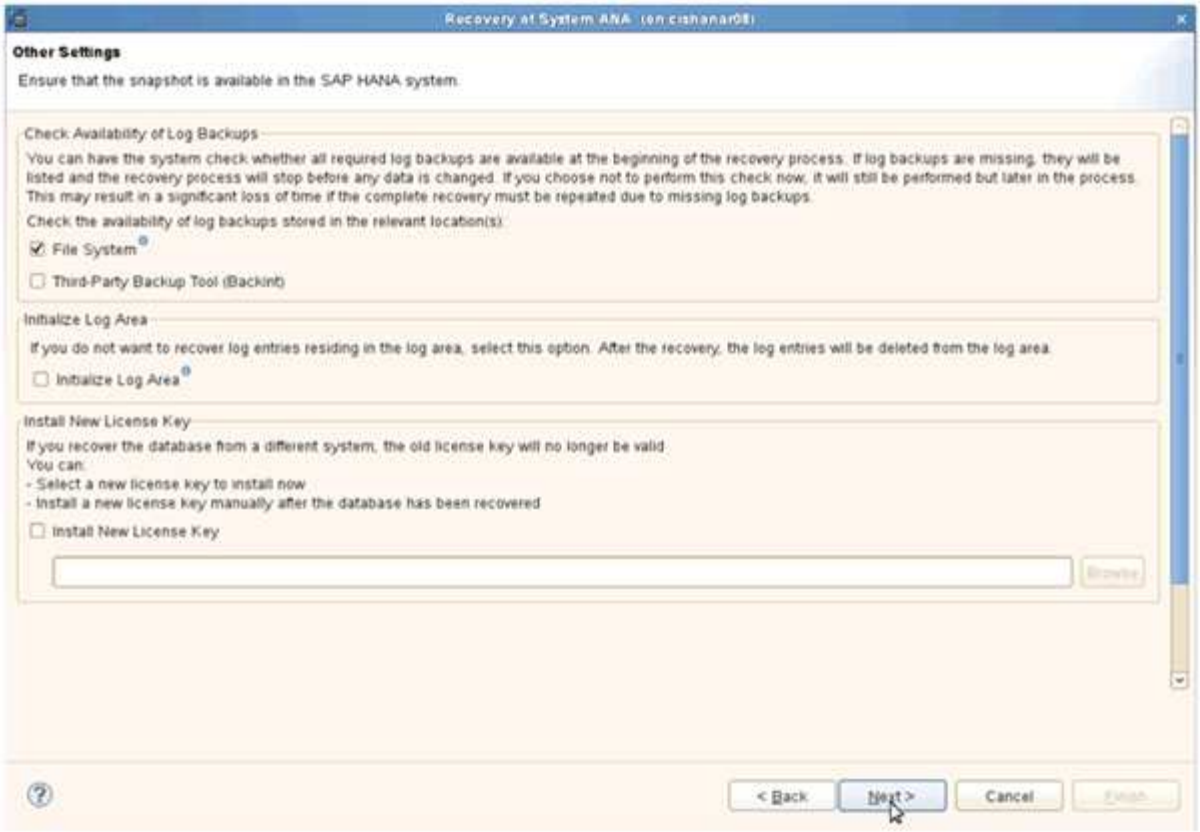
    在此示例中，所有三个卷都需要在每个数据库节点上重新挂载。

```
mount -o remount /hana/data/ANA/mnt00001
mount -o remount /hana/data/ANA/mnt00002
mount -o remount /hana/data/ANA/mnt00003
```

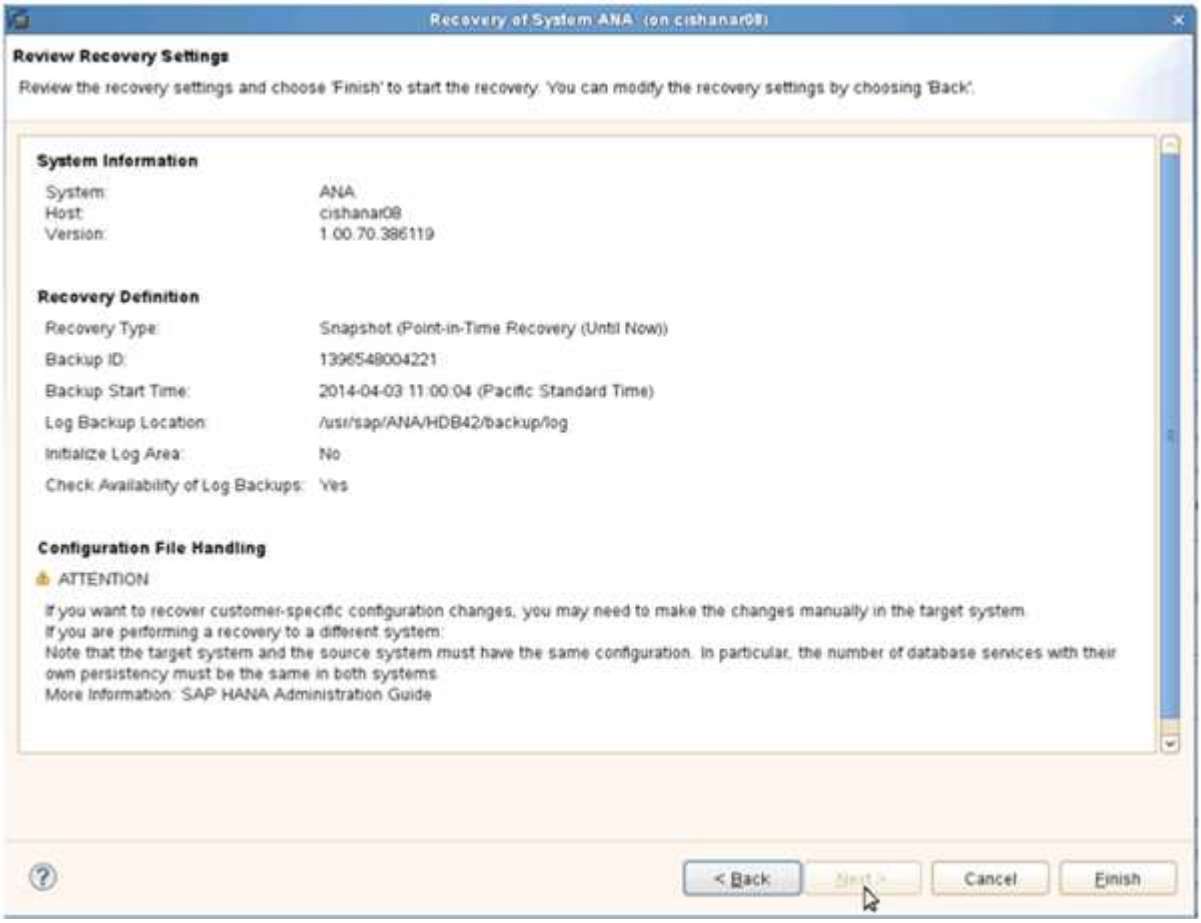
15. 转到 SAP HANA Studio 并单击 * 刷新 * 以更新备份列表。



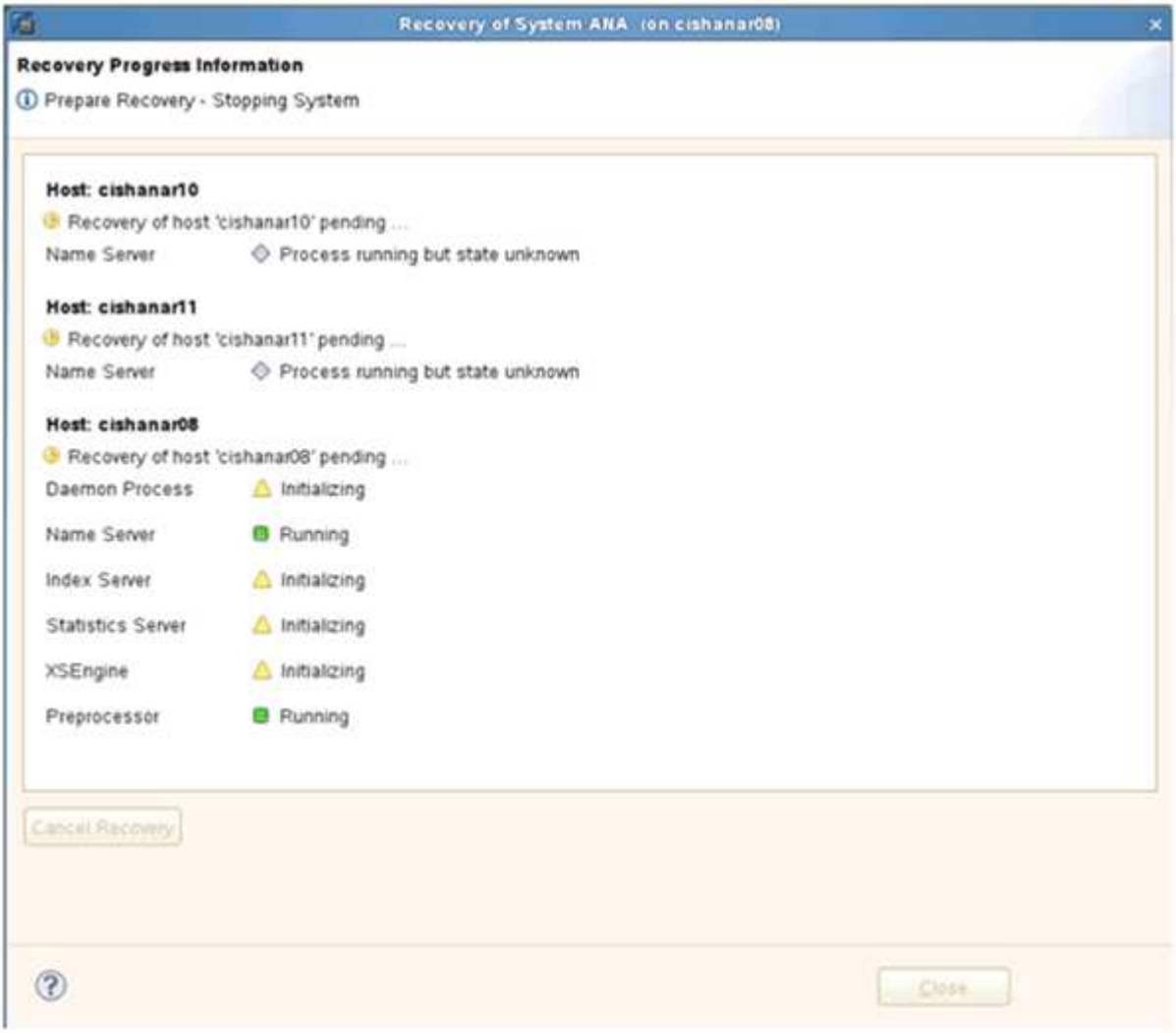
16. 使用 Snap Creator 还原的备份在备份列表中显示绿色图标。选择备份并单击 * 下一步 * 。
17. 根据需要选择其他设置，然后单击 * 下一步 * 。


Recovery of System ANA (on cishanar08)
Other Settings
Ensure that the snapshot is available in the SAP HANA system.
Check Availability of Log Backups
You can have the system check whether all required log backups are available at the beginning of the recovery process. If log backups are missing, they will be listed and the recovery process will stop before any data is changed. If you choose not to perform this check now, it will still be performed but later in the process. This may result in a significant loss of time if the complete recovery must be repeated due to missing log backups.
Check the availability of log backups stored in the relevant location(s):
File System
Third-Party Backup Tool (Backint)
Initialize Log Area
If you do not want to recover log entries residing in the log area, select this option. After the recovery, the log entries will be deleted from the log area.
Initialize Log Area
Install New License Key
If you recover the database from a different system, the old license key will no longer be valid.
You can:
- Select a new license key to install now
- Install a new license key manually after the database has been recovered
Install New License Key
Browse...
< Back
Next >
Cancel
Finish

18. 单击 * 完成 * 。


Recovery of System ANA (on cishanar08)
Review Recovery Settings
Review the recovery settings and choose 'Finish' to start the recovery. You can modify the recovery settings by choosing 'Back'.
System Information
System: ANA
Host: cishanar08
Version: 1.00.70.386119
Recovery Definition
Recovery Type: Snapshot (Point-in-Time Recovery (Until Now))
Backup ID: 1396548004221
Backup Start Time: 2014-04-03 11:00:04 (Pacific Standard Time)
Log Backup Location: /usr/sap/ANA/HDB42/backup/log
Initialize Log Area: No
Check Availability of Log Backups: Yes
Configuration File Handling
ATTENTION
If you want to recover customer-specific configuration changes, you may need to make the changes manually in the target system.
If you are performing a recovery to a different system:
Note that the target system and the source system must have the same configuration. In particular, the number of database services with their own persistency must be the same in both systems.
More Information: SAP HANA Administration Guide
< Back
Next >
Cancel
Finish

恢复过程开始。



19. 恢复过程完成后，根据需要恢复 SnapVault 关系。

还原后恢复 SnapVault 关系

如果未使用最新的 Snapshot 备份执行任何还原，则会删除主存储系统上的 SnapVault 关

系。

还原和恢复过程完成后，必须恢复 SnapVault 关系，以便可以使用 Snap Creator 再次执行备份。否则， Snap Creator 将问题描述显示一条错误消息，因为它在主存储系统上找不到 SnapVault 关系。

如果源卷和目标卷之间仍存在通用 Snapshot 副本，则所需的数据传输将基于增量传输。

恢复与 7- 模式 Data ONTAP 的 SnapVault 关系

如果使用最新的 Snapshot 备份以外的 Snapshot 备份进行还原，则需要恢复 SnapVault 关系，以便 Snap Creator 可以继续运行备份。

1. 输入以下命令，恢复与 7- 模式 Data ONTAP 的 SnapVault 关系。SnapVault start -r -S source_controller： source_volumebackup_controller： backup_volume

   对属于 SAP HANA 数据库的所有卷执行此步骤。

```
hana2b> snapvault start -r -S hana1a:/vol/data_00001/mnt00001
hana2b:/vol/backup_data_00001/mnt00001
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:08:21 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00001/mnt00001 to hana1a:/vol/data_00001/mnt00001 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana2b> snapvault start -r -S hana1b:/vol/data_00002/mnt00002
hana2b:/vol/backup_data_00002/mnt00002
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:09:49 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00002/mnt00002 to hana1b:/vol/data_00002/mnt00002 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana2b> snapvault start -r -S hana1a:/vol/data_00003/mnt00003
hana2b:/vol/backup_data_00003/mnt00003
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:10:25 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00003/mnt00003 to hana1a:/vol/data_00003/mnt00003 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

数据传输完成后，您可以再次使用 Snap Creator 计划备份。

恢复与集群模式 **Data ONTAP** 的 **SnapVault** 关系

如果使用最新的 Snapshot 备份以外的 Snapshot 备份进行还原，则需要恢复 SnapVault 关系，以便 Snap Creator 可以继续运行备份。

1. 重新创建并重新同步 SnapVault 关系。

```
hana::> snapmirror create -source-path hana1a:hana_data -destination
-path
hana2b:backup_hana_data -type XDP
Operation succeeded: snapmirror create the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror resync -destination-path hana2b:backup_hana_data
-type XDP

Warning: All data newer than Snapshot copy sc-backup-
daily_20140430121000 on volume
hana2b:backup_hana_data will be deleted.
Do you want to continue? {y|n}: y
[Job 6554] Job is queued: initiate snapmirror resync to destination
"hana2b:backup_hana_data".
[Job 6554] Job succeeded: SnapMirror Resync Transfer Queued
```

2. 要实际重新启动 SnapVault 传输，需要手动创建 Snapshot 副本。

```
hana::> snapshot create -vserver hana1a -volume hana_data -snapshot
sv_resync

hana::> snapshot modify -vserver hana1a -volume hana_data -snapshot
sv_resync -snapmirror-label daily

hana::> snapmirror update -destination-path hana2b:backup_hana_data
Operation is queued: snapmirror update of destination
hana2b:backup_hana_data.
```

3. 验证 SnapVault 关系是否显示在目标列表中。

```
hana::> snapmirror list-destinations -source-path hana1a:hana_data
                                                      Progress
Source              Destination         Transfer   Last
Relationship
Path         Type  Path         Status Progress   Updated      Id
----------- ----- ------------ ------- --------- ------------
---------------
hana1a:hana_data
            XDP   hana2b:backup_hana_data
                                Transferring
                                        38.46KB   04/30 18:15:54
                                                                 9137fb83-
cba9-11e3-85d7-123478563412
```

在主存储发生故障后还原数据库

在主存储发生故障或从主存储的卷中删除所有 Snapshot 副本后， Snap Creator 将无法处理还原，因为主存储系统上将不再存在 SnapVault 关系。

使用在 **7-** 模式下运行的 **Data ONTAP** 在主存储发生故障后还原数据库

在运行 7- 模式 Data ONTAP 的主存储系统出现故障后，您可以还原 SAP HANA 数据库。

1. 在这种情况下，必须使用以下命令在二级存储系统上直接执行还原： SnapVault restore -s snapshot_name -S backup_controller： backup_volumesource_controller： source_volume

   对属于 SAP HANA 数据库的所有卷执行此步骤。

```
hana1a>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00001/mnt00001 hana1a:/vol/data_00001/mnt00001
Restore will overwrite existing data in /vol/data_00001/mnt00001.
Are you sure you want to continue? y
Thu Apr 10 11:55:55 CEST [hana1a:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00001/mnt00001.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana1a>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00003/mnt00003 hana1a:/vol/data_00003/mnt00003
Restore will overwrite existing data in /vol/data_00003/mnt00003.
Are you sure you want to continue? y
Thu Apr 10 11:58:18 CEST [hana1a:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00003/mnt00003.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana1b>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00002/mnt00002 hana1b:/vol/data_00002/mnt00002
Restore will overwrite existing data in /vol/data_00002/mnt00002.
Are you sure you want to continue? y
Thu Apr 10 12:01:29 CEST [hana1b:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00002/mnt00002.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

还原过程完成后，您可以使用 SAP HANA 执行恢复。

使用集群模式 **Data ONTAP** 在主存储发生故障后还原数据库

您可以在运行集群模式 Data ONTAP 的主存储系统出现故障后还原 SAP HANA 数据库。

假设主卷完全丢失，您需要创建一个新的主卷，然后从备份卷进行还原。

1. 创建类型为数据保护的主卷。

```
hana::> volume create -vserver hana1a -volume hana_data -aggregate
aggr_sas_101 -size 300G -state online -type DP -policy default -autosize
-mode grow_shrink -space-guarantee none
-snapshot-policy none -foreground true
[Job 6744] Job is queued: Create hana_data.
[Job 6744] Job succeeded: Successful
```

2. 从备份卷还原所有数据。

```
hana::> snapmirror restore -destination-path hana1a:hana_data -source
-path hana2b:backup_hana_data -source-snapshot sc-backup-
daily_20140505121000
[Job 6746] Job is queued: snapmirror restore from source
"hana2b:backup_hana_data" for the
snapshot sc-backup-daily_20140505121000.

hana::> job show -id 6746
Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
6746   SnapMirror restore   hana        hana01        Running
       Description: snapmirror restore from source
"hana2b:backup_hana_data" for the snapshot sc-backup-
daily_20140505121000
```

还原过程完成后，您可以使用 SAP HANA 执行恢复。

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